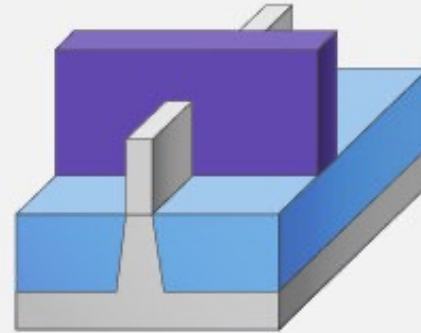
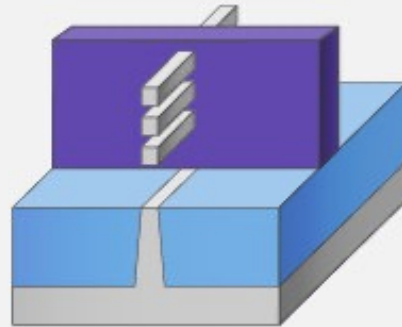


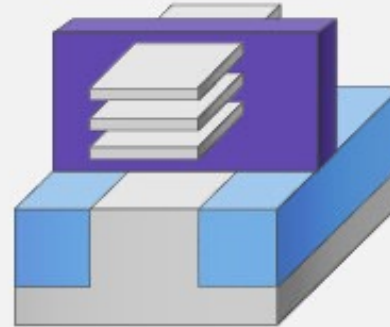
Planar FET



FinFET



GAAFET
(Nanowire)



MBCFET™
(Nanosheet)

da Jean-Luc Aufranc (CNXSoft)

Evoluzione dei Transistori MOS

Evoluzione dei Transistori MOS

Corso di Microelettronica

Laurea Magistrale in Ingegneria Elettronica

Dipartimento di Ingegneria dell'Informazione - Università degli Studi di Brescia

Zsolt M. Kovács - Vajna

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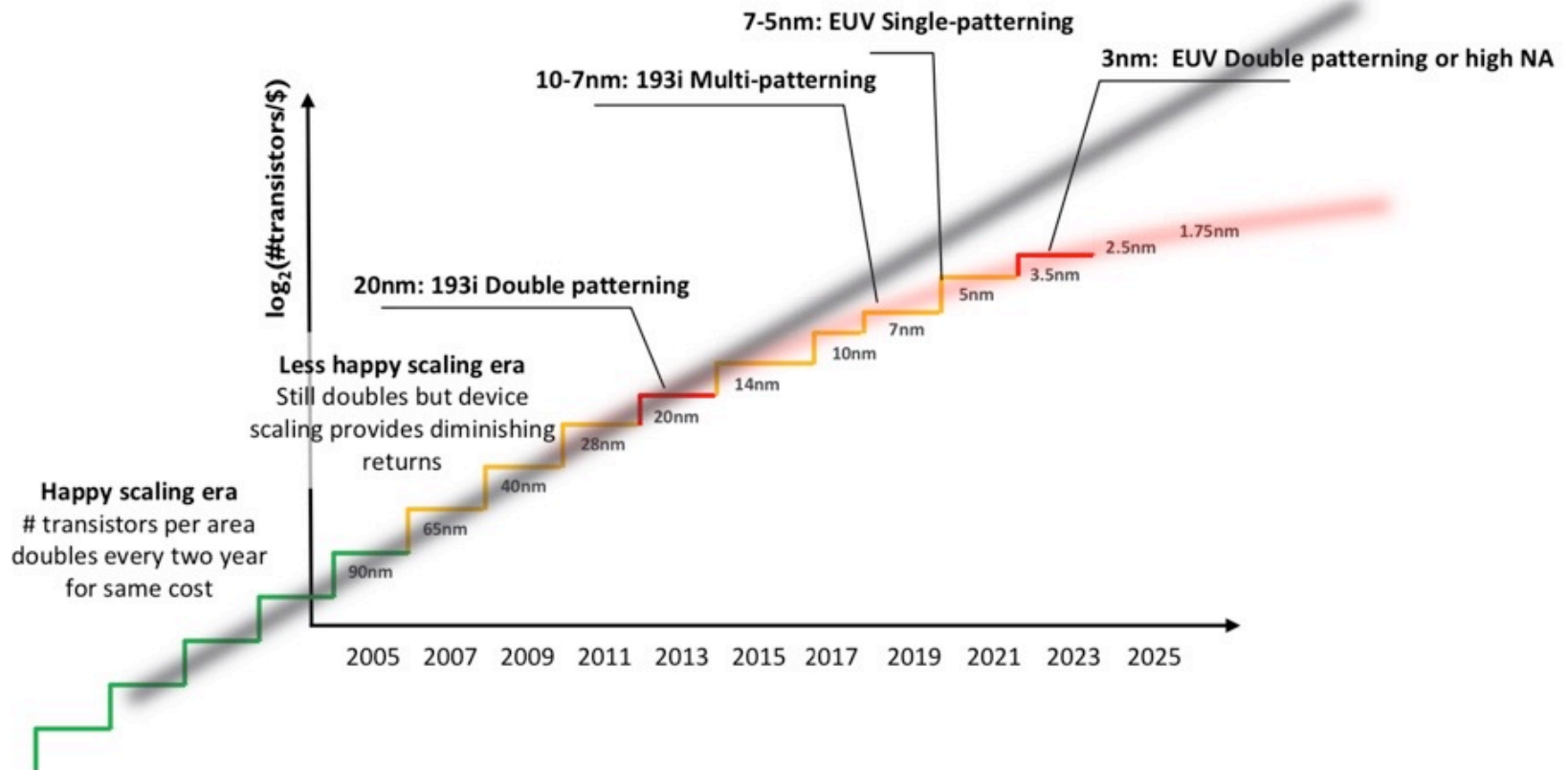
How Microchips are made

8:53

<https://youtu.be/F2KcZGwntgg>

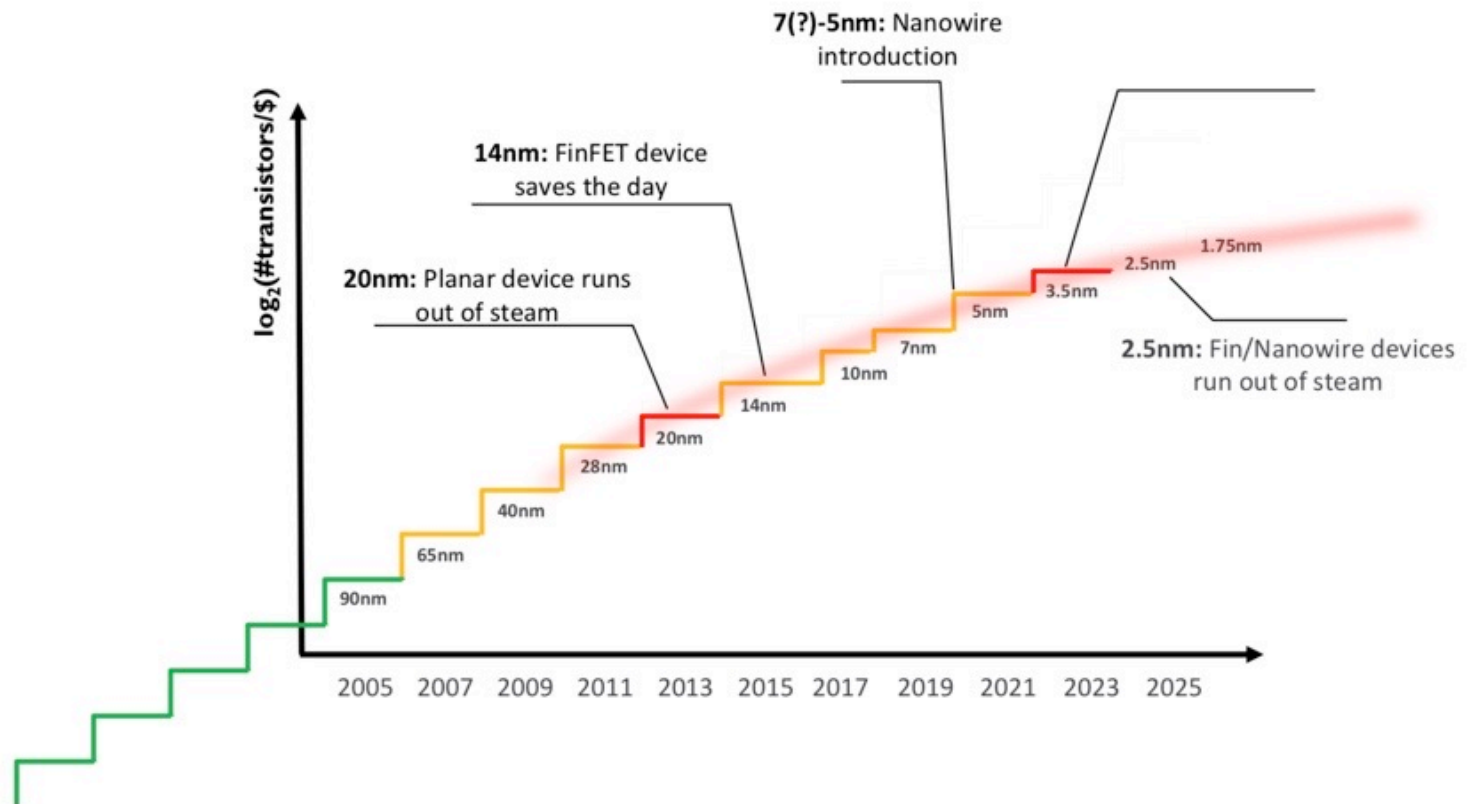


Dimensional scaling under pressure



Technology Options at the 5-Nanometer Node

Transistor architecture under pressure



Technology Options at the 5-Nanometer Node

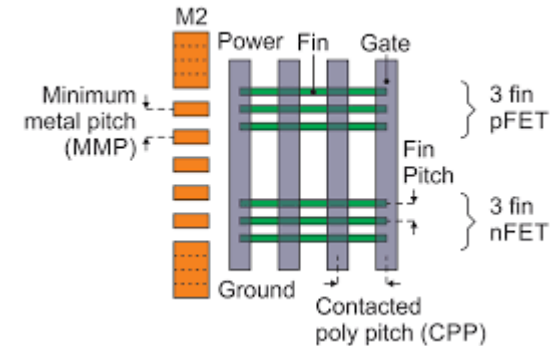
INIZIALE EVOLUZIONE TECNOLOGICA

<i>Year of Introduction</i>	<i>1994</i>	<i>1997</i>	<i>2000</i>	<i>2003</i>	<i>2006</i>	<i>2009</i>
<i>Channel length (μm)</i>	0.4	0.3	0.25	0.18	0.13	0.1
<i>Gate oxide (nm)</i>	12	7	6	4.5	4	4
<i>V_{DD} (V)</i>	3.3	2.2	2.2	1.5	1.5	1.5
<i>V_T (V)</i>	0.7	0.7	0.7	0.6	0.6	0.6
<i>NMOS I_{Dsat} (mA/μm) (@ $V_{GS} = V_{DD}$)</i>	0.35	0.27	0.31	0.21	0.29	0.33
<i>PMOS I_{Dsat} (mA/μm) (@ $V_{GS} = V_{DD}$)</i>	0.16	0.11	0.14	0.09	0.13	0.16

ATTUALE EVOLUZIONE TECNOLOGICA

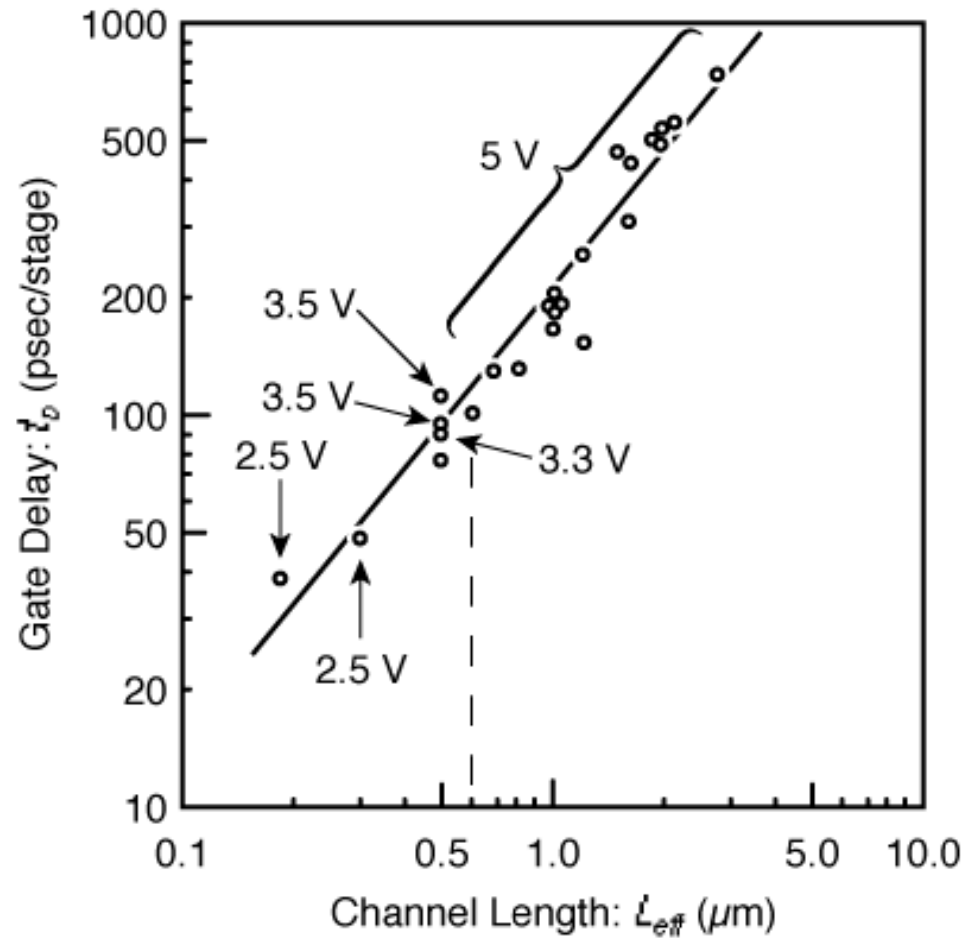
IEEE Advancing Technology for Humanity

Device architecture and ground rules roadmap for logic device technologies. PxxMxx notation refers to Pxx: contacted poly pitch and Mxx: metalx pitch in nm.



YEAR OF PRODUCTION	2015	2017	2019	2021	2024	2027	2030
Logic device technology naming	P70M56	P54M36	P42M24	P32M20	P24M12G1	P24M12G2	P24M12G3
Logic industry "Node Range" Labeling (nm)	"16/14"	"11/10"	"8/7"	"6/5"	"4/3"	"3/2.5"	"2/1.5"
Logic device structure options	finFET FDSOI	finFET FDSOI	finFET LGAA	finFET LGAA VGAA	VGAA, M3D	VGAA, M3D	VGAA, M3D
DEVICE PHYSICAL&ELECTRICAL SPECS							
Power Supply Voltage - V_{dd} (V)	0.80	0.75	0.70	0.65	0.55	0.45	0.40
Subthreshold slope - [mV/dec]	75	70	68	65	40	25	25
Inversion layer thickness - [nm] [4]	1.10	1.00	0.90	0.85	0.80	0.80	0.80
$V_{t,sat}$ (mV) at $I_{off}=100nA/\mu m$ - HP Logic [5] [6]	129	129	133	136	84	52	52
$V_{t,sat}$ (mV) at $I_{off}=100pA/\mu m$ - LP Logic [5] [6]	351	336	333	326	201	125	125
Effective mobility ($cm^2/V.s$)	200	150	120	100	100	100	100

EFFETTO: es. TEMPO DI PROPAGAZIONE



- ***Full Scaling*** (a **campo elettrico costante**): vengono scalate le geometrie, profondità di giunzione e tensioni di un fattore ‘S’. (V_{DD} spesso fissato; Φ_T , Φ_B , Φ_F , E_G non si scalano; V_T limita la conduzione sotto-soglia)
- ***General Scaling*** (geometrie e tensione di alimentazione scalano in modo indipendente): le geometrie vengono scalate di un fattore ‘S’, la tensione di alimentazione di ‘U’. (meno vantaggioso del ***Full Scaling***, se $U < S$)
- ***Fixed-Voltage Scaling*** (tensione di alimentazione costante: $U = 1$): le geometrie vengono scalate di un fattore ‘S’. (realizzabile e fatto fino ad ora quasi sempre)

SCALING DEI PARAMETRI

Parameter	Relation	Full Scaling	General Scaling	Fixed Voltage Scaling	MOS a canale lungo $\mu = \text{cost}$
W, L, t_{ox}		1/S	1/S	1/S	
V_{DD}, V_T		1/S	1/U	1	
N_{SUB}	V/W_{depl}^2	S	S^2/U	S^2	$\leftarrow X_j$
Area/ Device	WL	$1/S^2$	$1/S^2$	$1/S^2$	
C_{ox}	$1/t_{ox}$	S	S	S	
C_L	$C_{ox} WL$	1/S	1/S	1/S	
k_n, k_p	$C_{ox} W/L$	S	S	S	
I_{av}	$k_{n,p} V^2$	1/S	S/U^2	S	
t_p (intrinsic)	$C_L V/I_{av}$	1/S	U/S^2	$1/S^2$	
P_{av}	$C_L V^2/t_p$	$1/S^2$	S/U^3	S	
PDP	$C_L V^2$	$1/S^3$	$1/S U^2$	1/S	

Nei dispositivi submicrometrici non sono trascurabili alcuni effetti che sono del secondo ordine nei MOS a canale lungo:

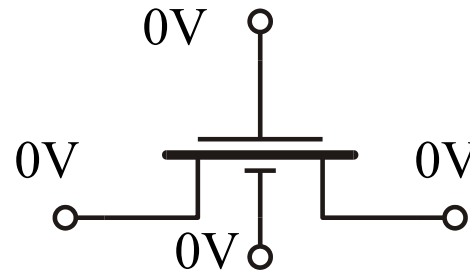
- ✓ Degradazione della mobilità
- ✓ Saturazione della velocità
- ✓ ***DIBL*** (*Drain Induced Barrier Lowering*)
- ✓ Resistenze serie

SCALING DEI PARAMETRI

Parameter	Relation	Full Scaling	General Scaling	Fixed Voltage Scaling	MOS a canale corto $\mu = \text{cost}$
W, L, t_{ox}		1/S	1/S	1/S	
V_{DD}, V_T		1/S	1/U	1	
N_{SUB}	V/W_{depl}^2	S	S^2/U	S^2	$\leftarrow X_j$
Area/ Device	WL	$1/S^2$	$1/S^2$	$1/S^2$	
C_{ox}	$1/t_{ox}$	S	S	S	
C_L	$C_{ox} WL$	1/S	1/S	1/S	
k_n, k_p	$C_{ox} W/L$	S	S	S	
I_{av}	$C_{ox} WV$	1/S	$1/U$ (S/U ²)	1 (S)	
t_p (intrinsic)	$C_L V/I_{av}$	1/S	$1/S$ (U /S ²)	$1/S$ (1/S ²)	
P_{av}	$C_L V^2 / t_p$	$1/S^2$	$1/U^2$ (S/U ³)	1 (S)	
PDP	$C_L V^2$	$1/S^3$	$1/S U^2$	1/S	

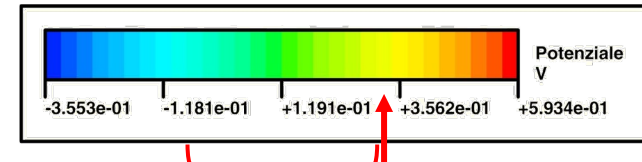
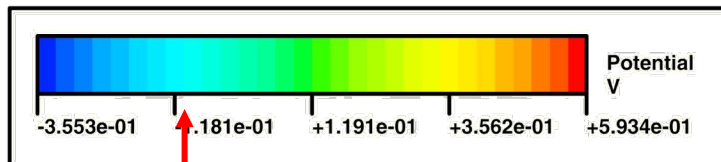
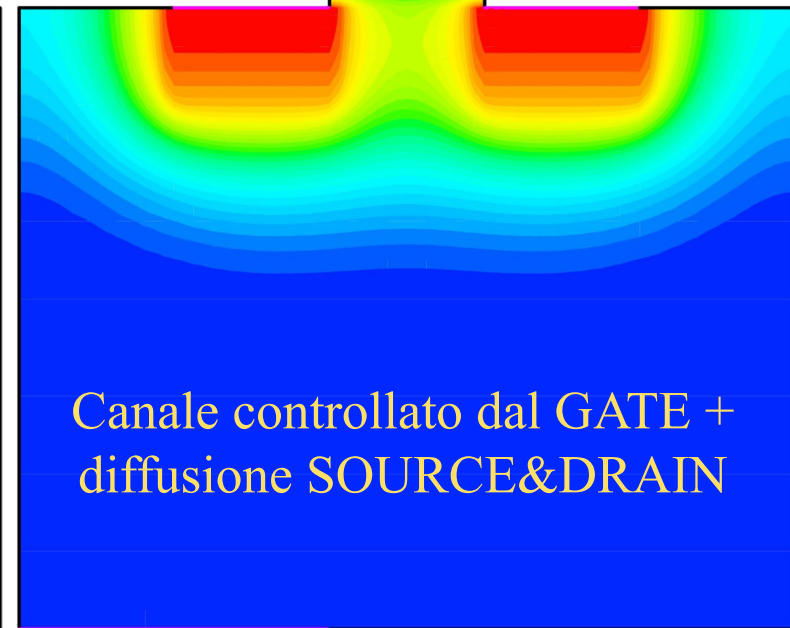
- **Corrente di OFF nei MOS**
- Limiti del Fattore di Conduzione k nei MOS
- Fotolitografia

MOS: problema della corrente di OFF



dispositivo a canale lungo

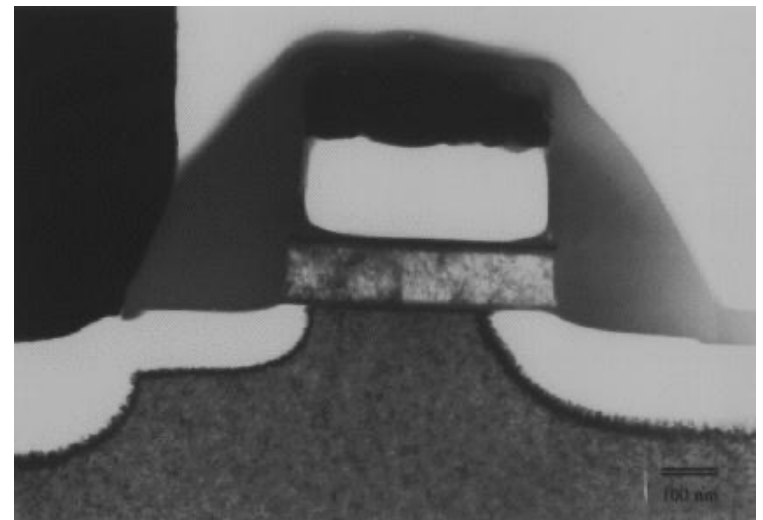
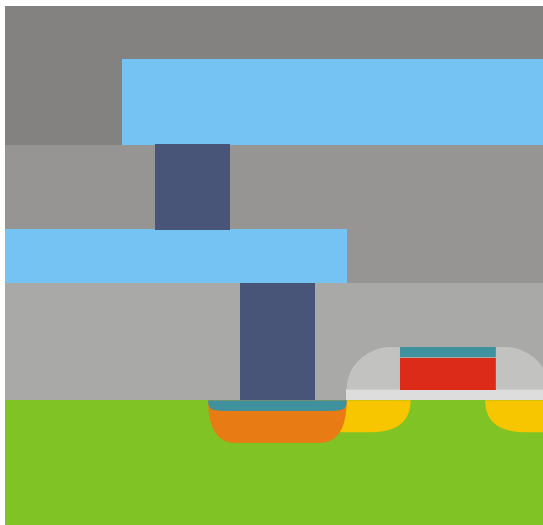
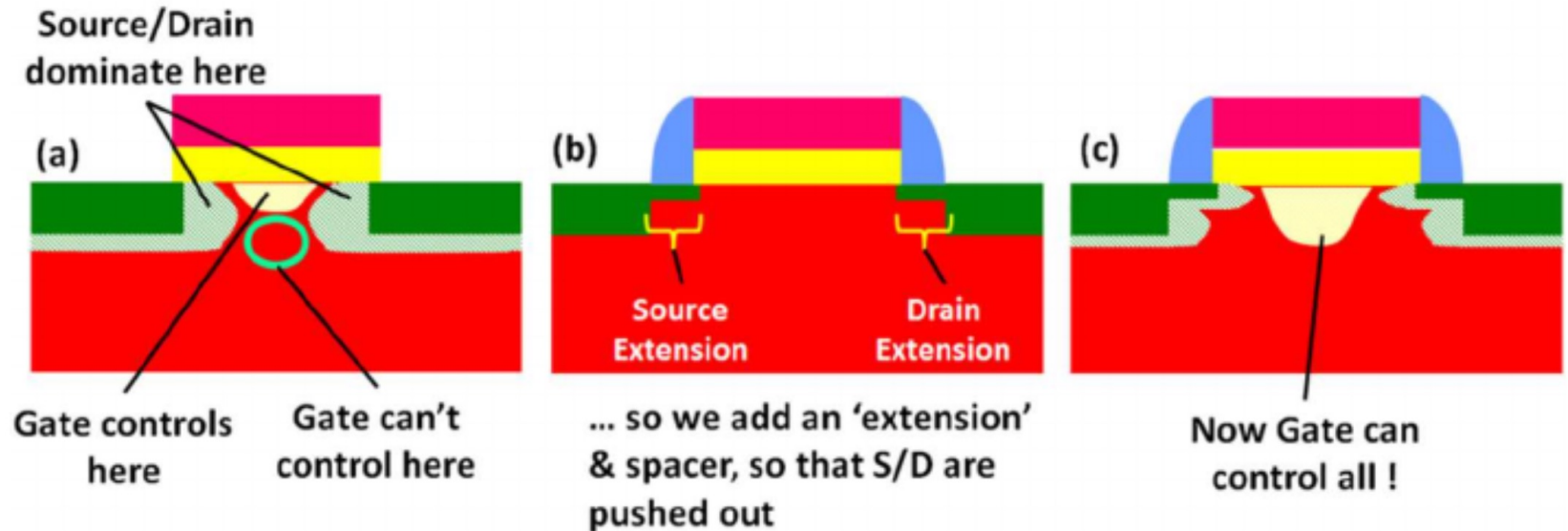
dispositivo a canale corto



Pre-Accensione del Canale: Corrente di OFF

Lightly Doped Drain (LDD) sotto 130nm

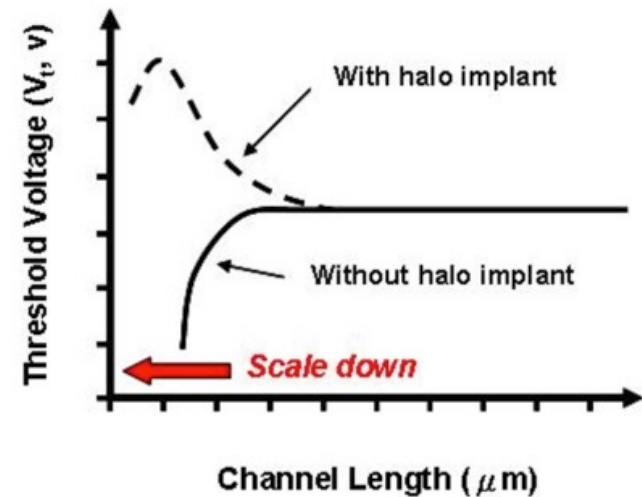
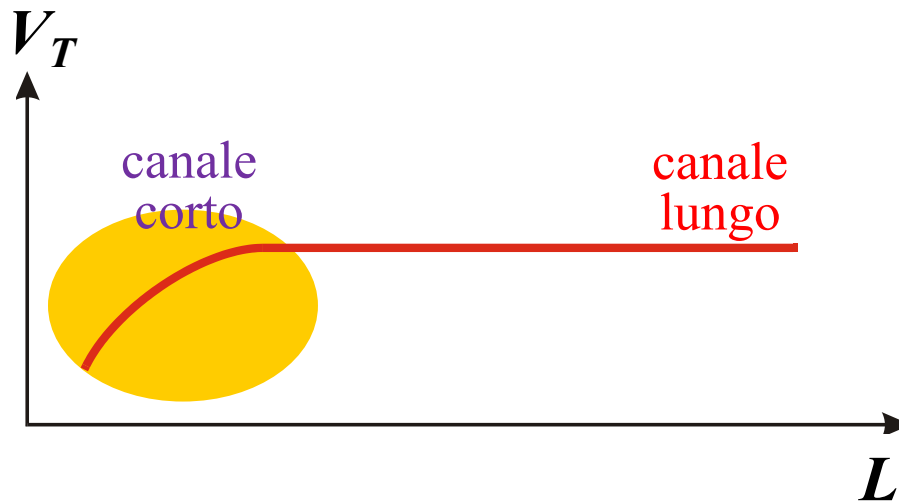
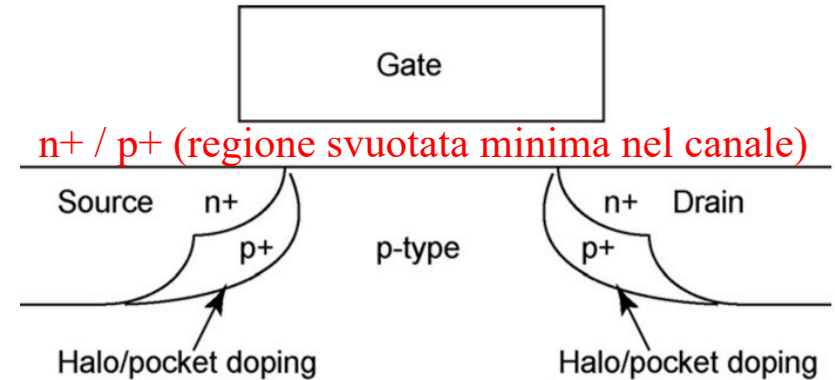
Ajey P. Jacob et al., "Scaling Challenges for Advanced CMOS Devices"



Reversed Short Channel Effect: *Halo Doping*

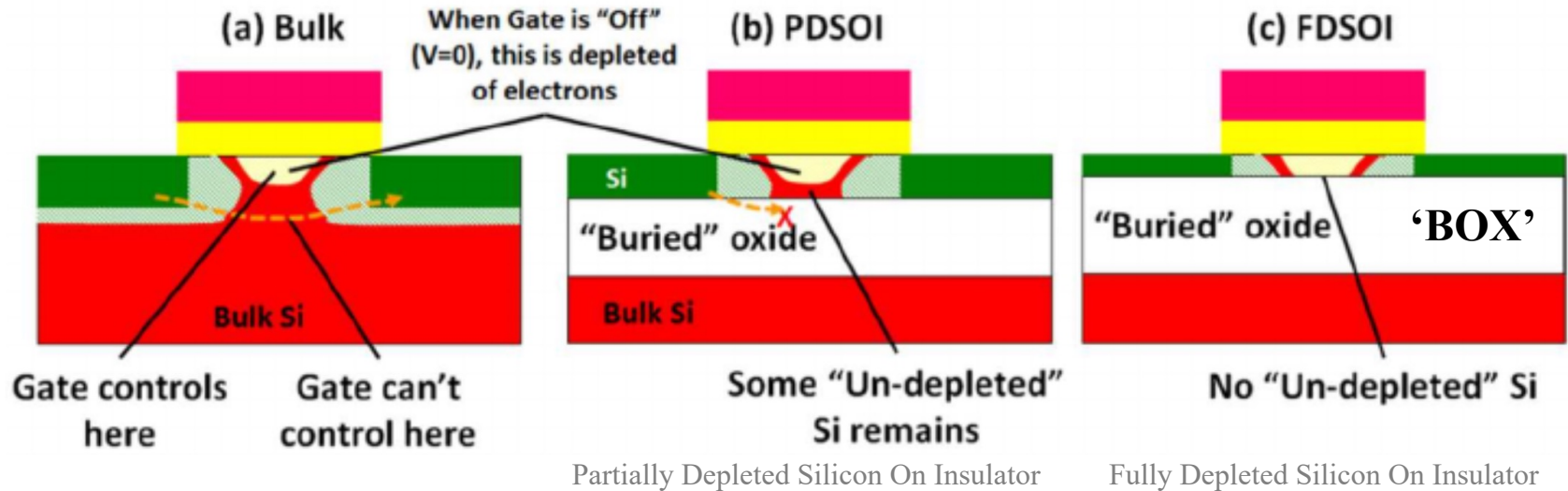
D/S n++ implica ampia regione
svuotata nel substrato p-:

n+ / p- (regione svuotata ampia nel canale)

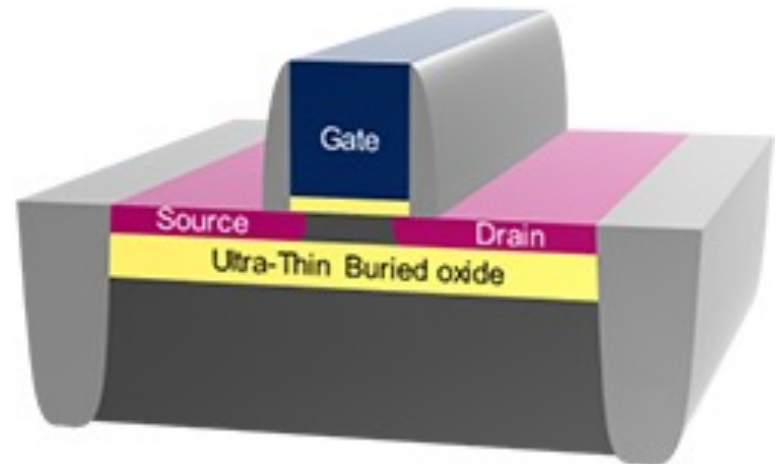


Controllo del Gate ($L > 25\text{nm}$)

Ajeay P. Jacob et al., "Scaling Challenges for Advanced CMOS Devices"



FD-SOI
(Fully Depleted Silicon On Insulator)
by STMicroelectronics



STM 28nm UTBB FD-SOI MOSFET

Ultra-Thin BODY

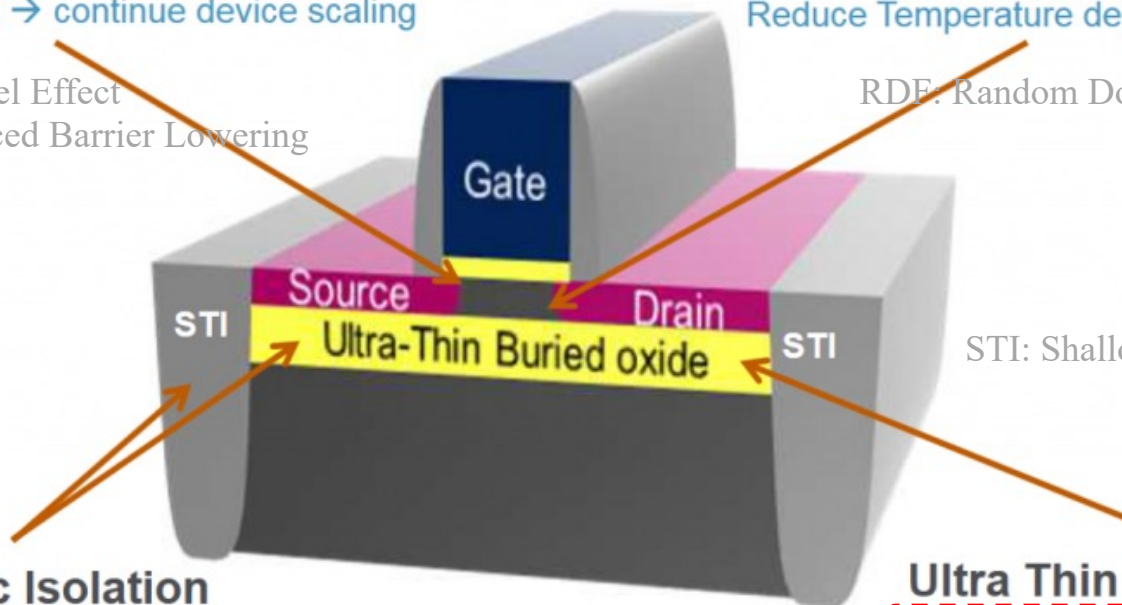
Excellent Electrostatic Control of the channel
Low SCE, DIBL $\rightarrow$ Low V_t @ High V_D
Gate length shrink $\rightarrow$ continue device scaling

SCE: Short Channel Effect
DIBL: Drain Induced Barrier Lowering

Undoped Channel & no Pocket

Less V_t variability & SRAM V_{min} improved
Lower Power Consumption
Reduce Temperature dependency & no RDF

RDF: Random Dopant Fluctuation



STI: Shallow Trench Isolation

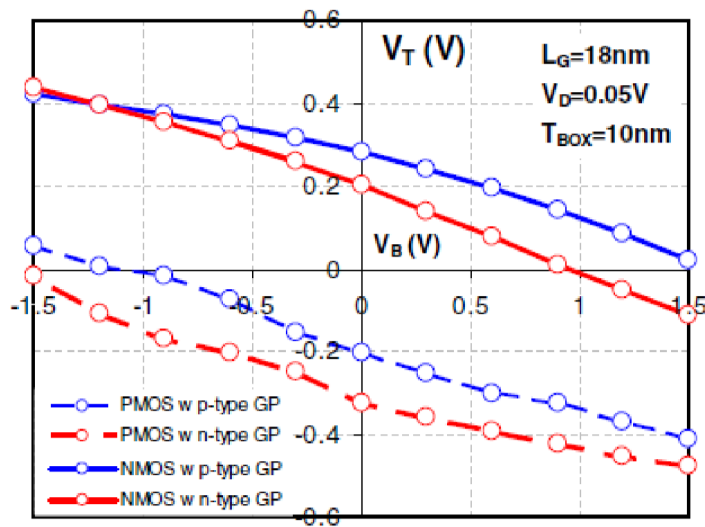
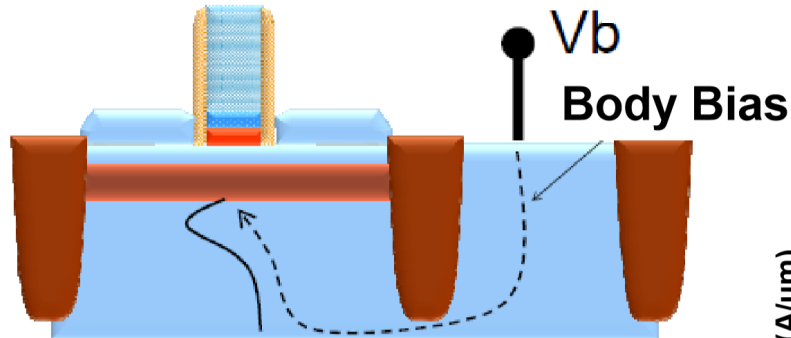
Total Dielectric Isolation

Lower SD capacitances & Lower SD Leakage
Less Sensitive to Temperature
Higher power efficiency

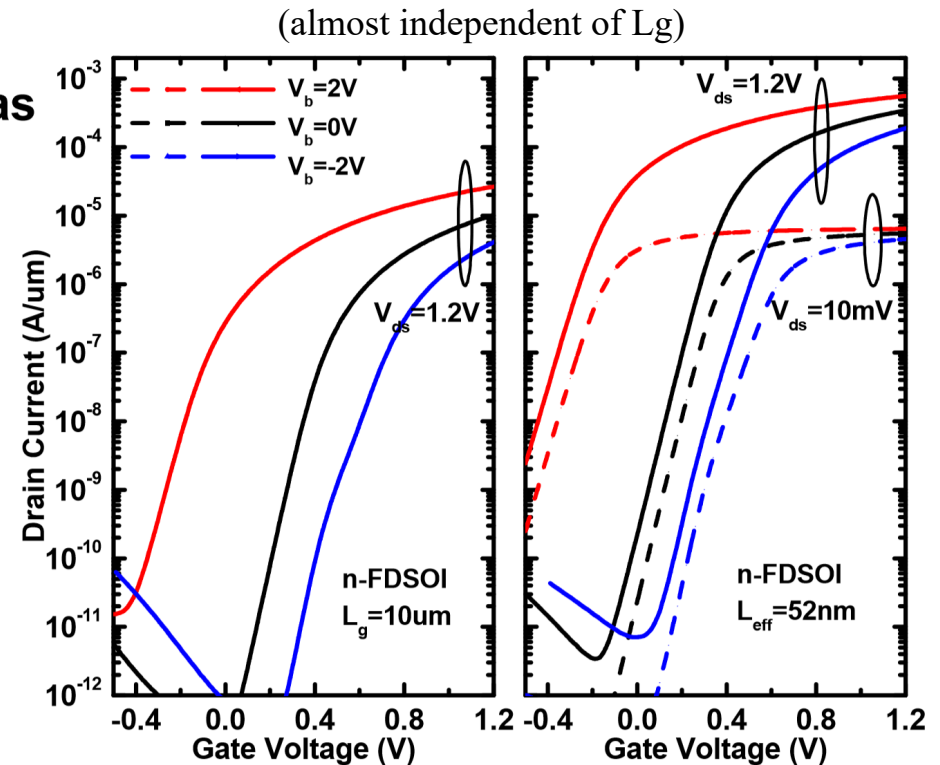
Ultra Thin BOX

Body Biasing (BB) $\rightarrow$ FBB & RBB
Speed boost due to BB
GP Implantation $\rightarrow$ V_t adjustment
GP: Ground Plane

Back Biasing Tuning in SOI MOSFETs

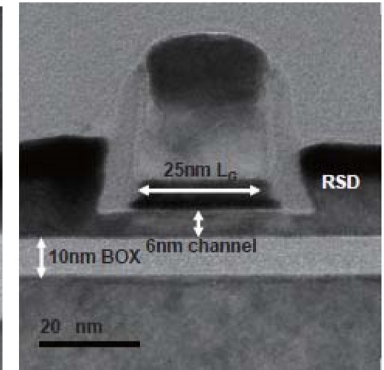
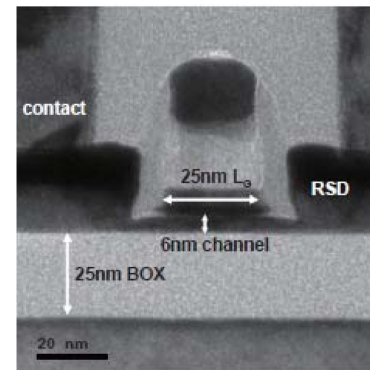
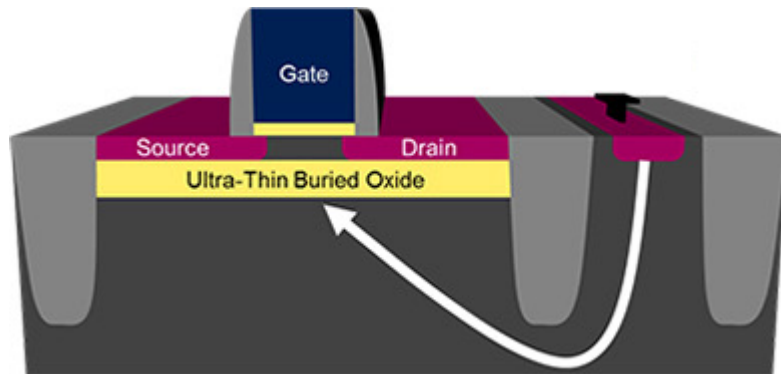


T. Skotnicki, IEDM SC (2010)



- Reverse Back Biasing:** (back surface more depleted)
 V_{BG} : NMOS(-), PMOS(+) $\rightarrow V_{TH} \uparrow$
- Forward Back Biasing:** (back surface less depleted)
 V_{BG} : NMOS(+), PMOS(-) $\rightarrow V_{TH} \downarrow$

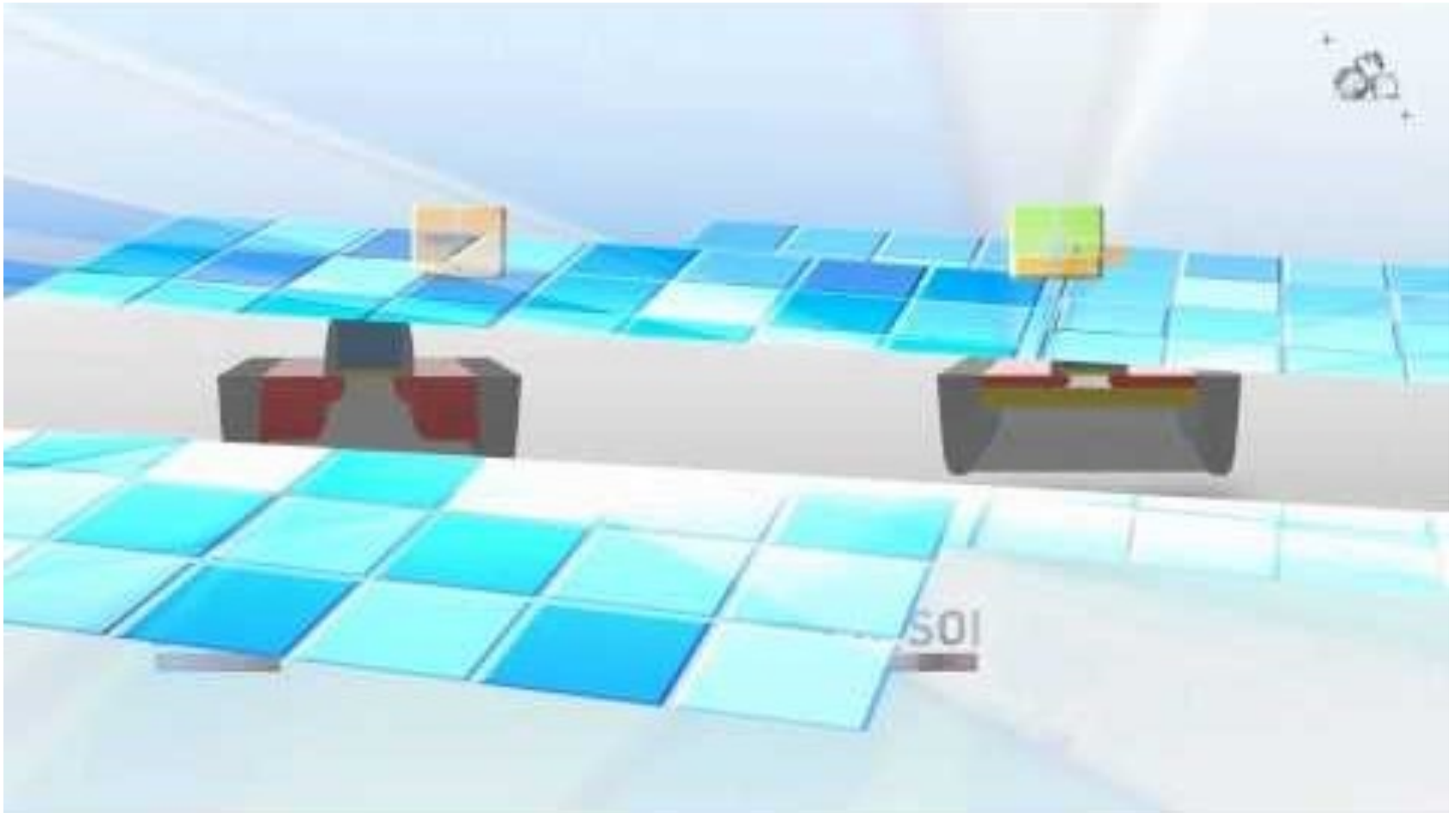
STM 28nm UTBB FD-SOI MOSFET



T_{BOX}	back bias	I_{on} ($\mu\text{A}/\mu\text{m}$)	I_{off} ($\text{nA}/\mu\text{m}$)	DIBL (mV/V)	SS (mV/dec)
25nm	-2V	525	0.1	68	90
	-1V	602	0.63	68	89
	0V	682	3.69	70	91
	1V	759	19.9	71	91
	2V	833	126	71	90
10nm	-2V	274	0.009	48	87
	-1V	441	0.26	56	89
	0V	628	10.5	64	91
	1V	807	316	78	93
	2V	958	5040	90	94

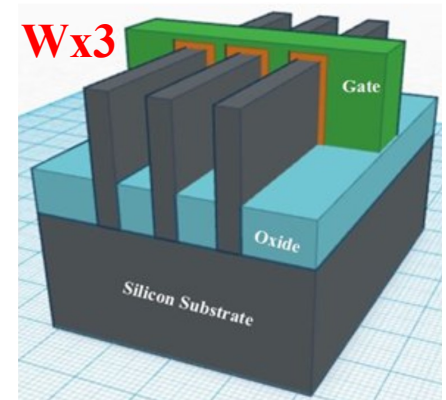
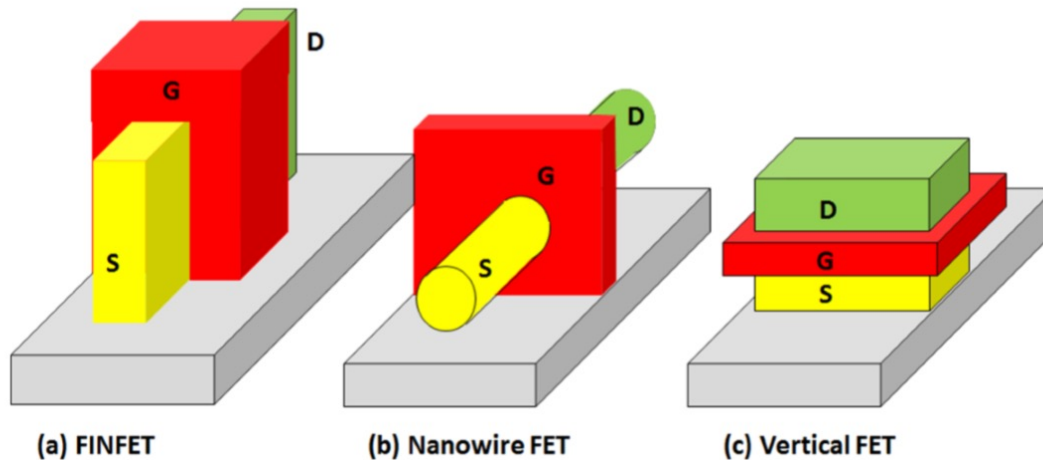
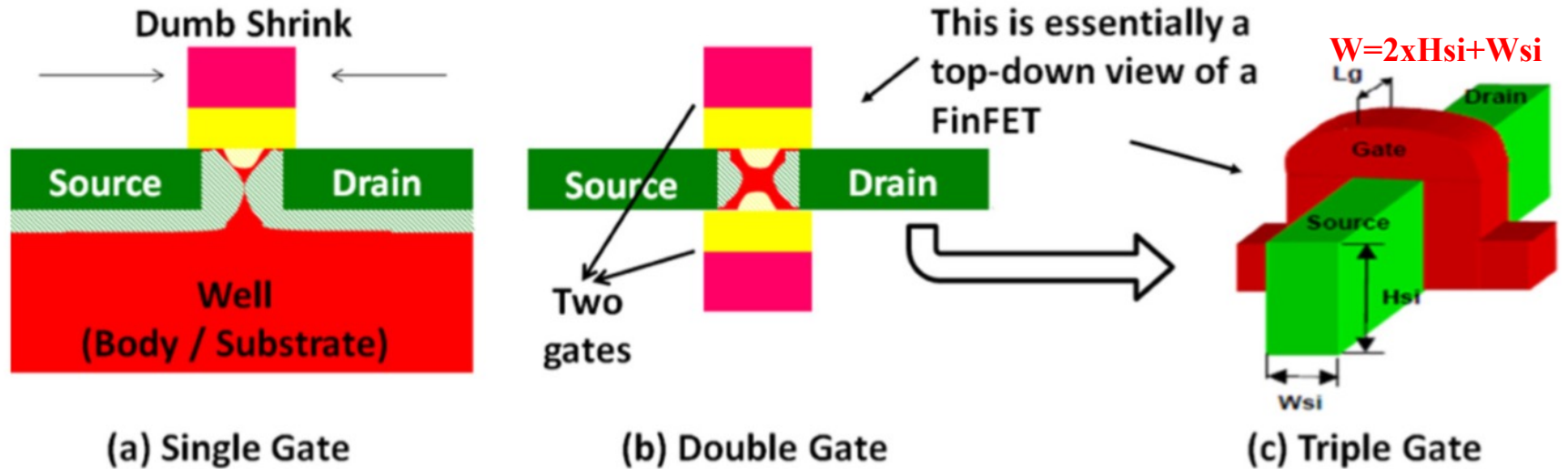
An introduction to FD-SOI

<https://youtu.be/uvV7jcpQ7UY>



Ulteriore riduzione di L (<25nm)

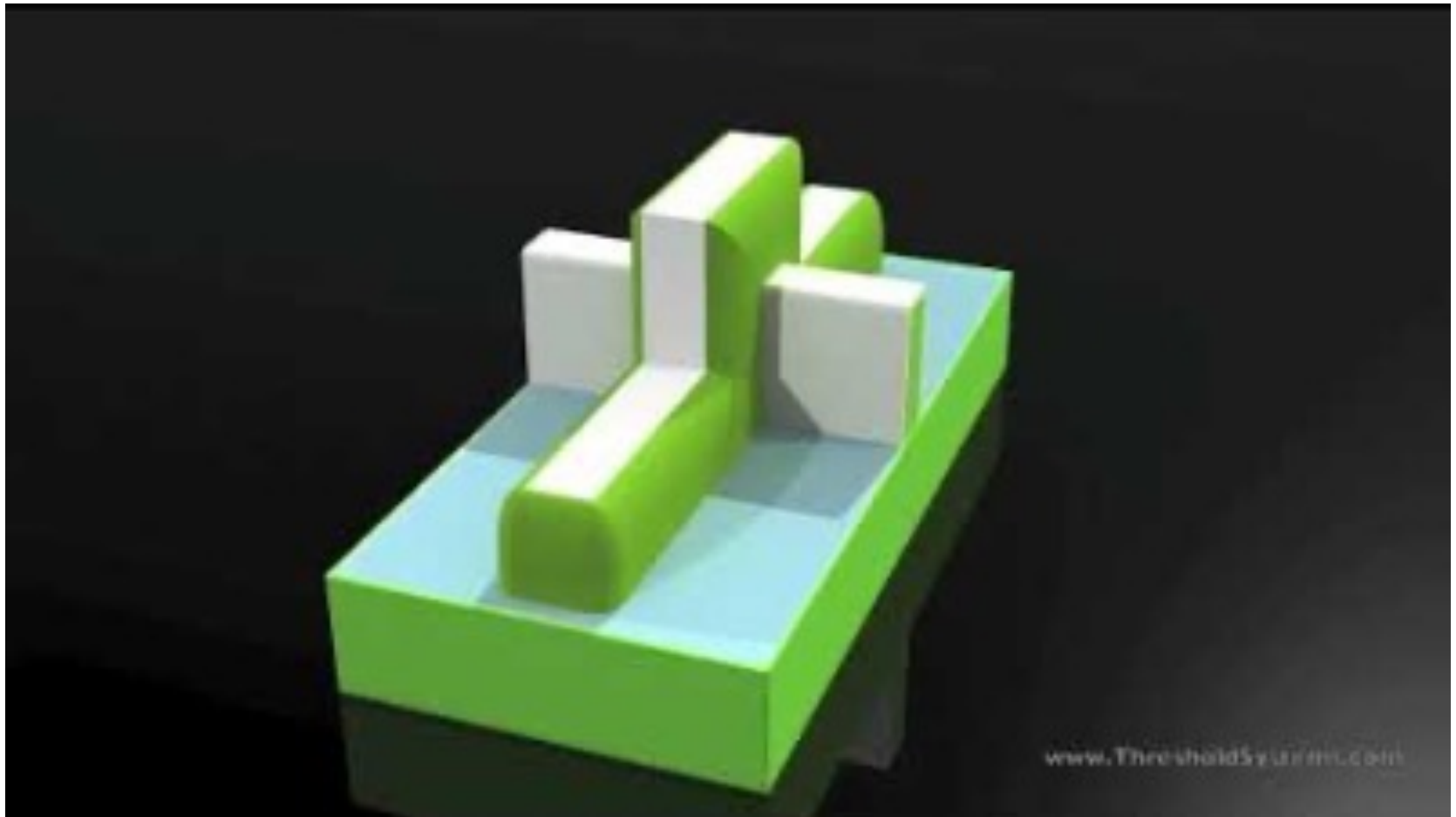
Ajey P. Jacob et al., "Scaling Challenges for Advanced CMOS Devices"



Understanding The FinFet Semiconductor Process

3:38

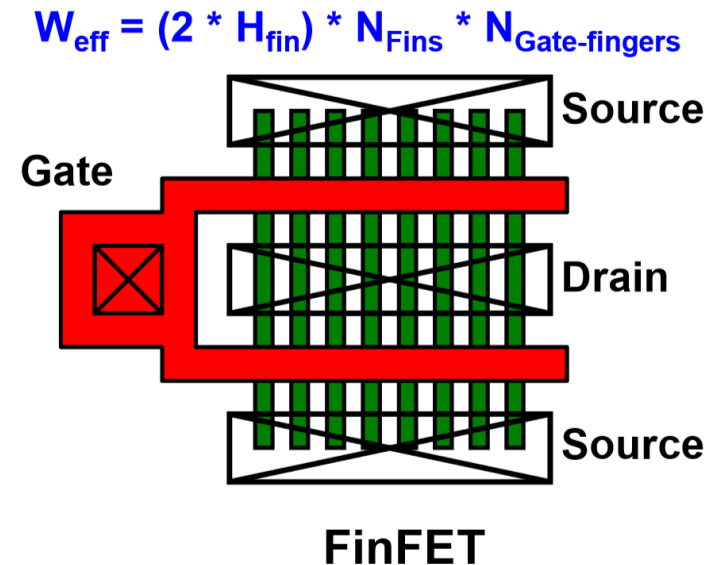
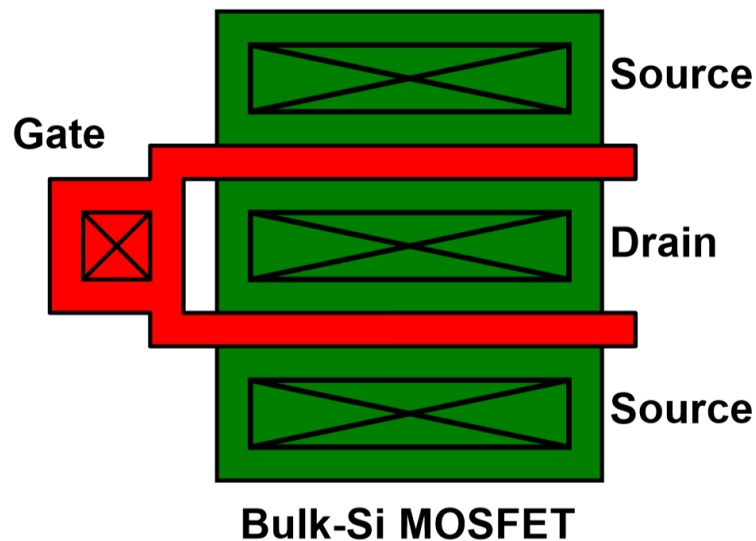
<https://www.youtube.com/watch?v=knkmlG6ItYo>



attenzione: W FinFet è mostrato in modo errato!

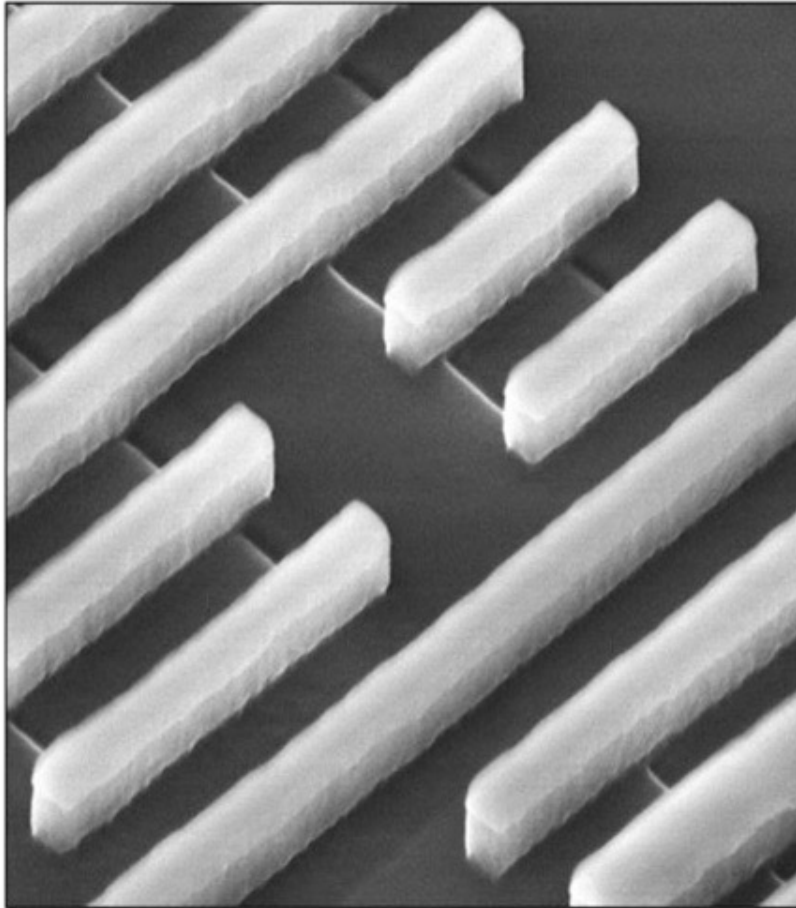
FinFET: L fisso, W quantizzato

Layout is similar to that of conventional MOSFET, except that the channel width is quantized:

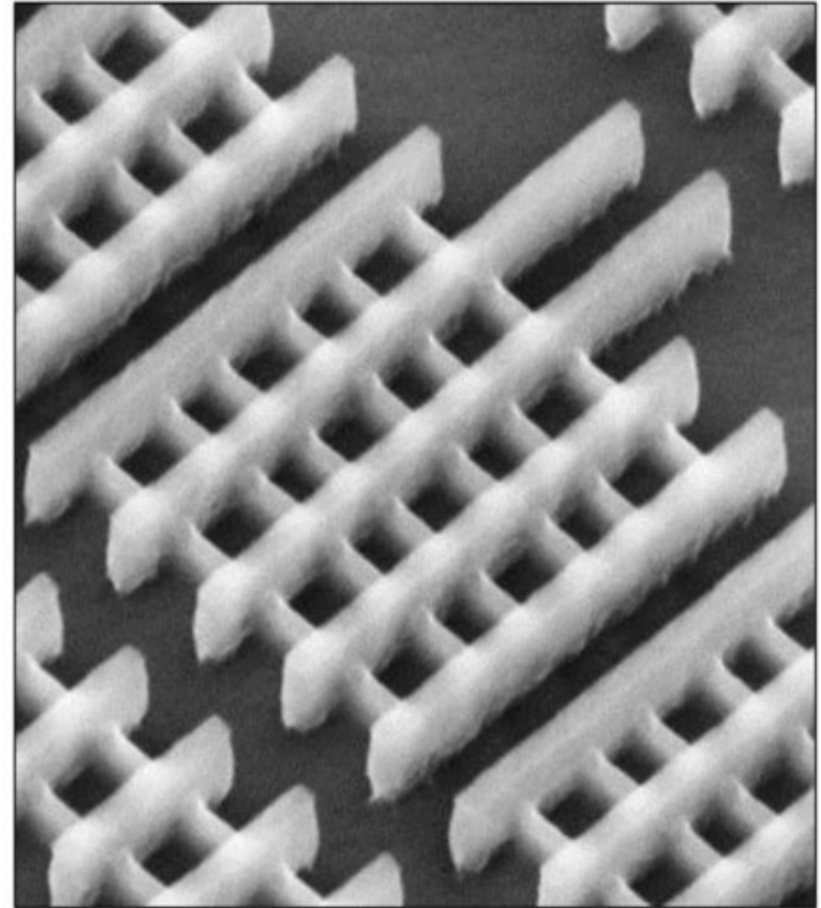


Planar vs. FinFET

32 nm Planar Transistors



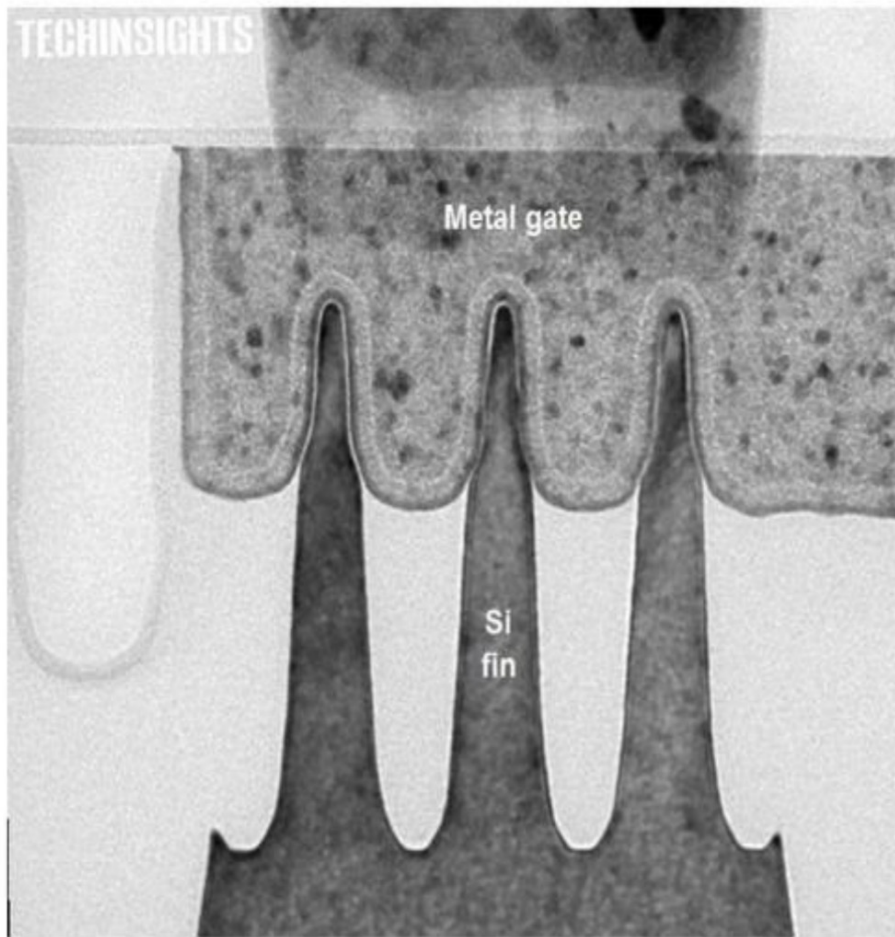
22 nm Tri-Gate Transistors



Mark Bohr , et al. (Intel) (2011)

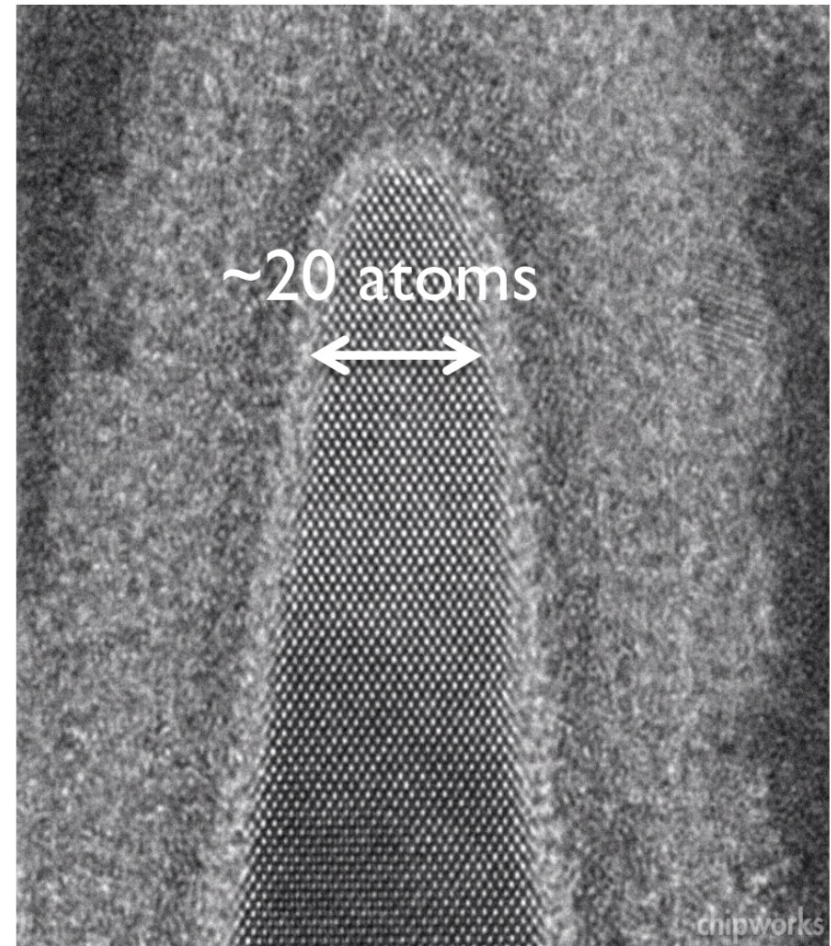
Fin Structure

Samsung 14nm Fins



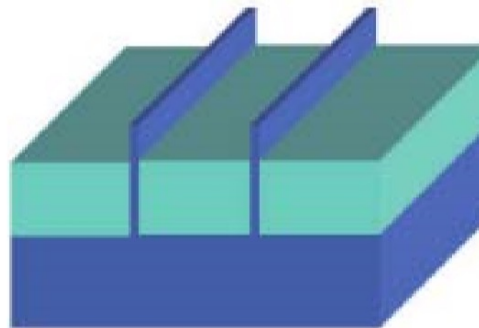
TechInsights (2015)

TEM Image of Intel 22nm Fin

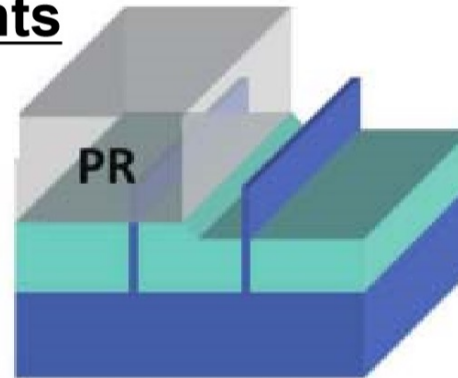


Chipworks (2012)

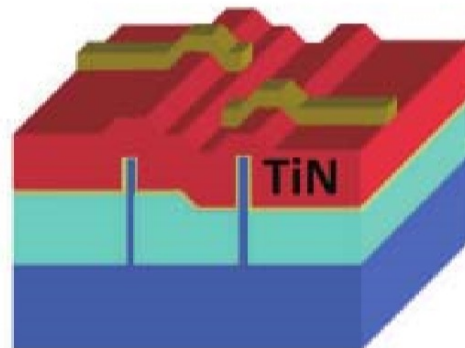
Process flow to form multiple fin heights



OX etching



**Recess mask
& OX recess**



**Gox + TiN gate dep.
& Gate patterning (NIL)**



Gate etching

11/17/2013

Nuo Xu

EE 290D, Fall 2013

- Corrente di OFF nei MOS
- **Limiti del Fattore di Conduzione k nei MOS**
- Fotolitografia

Fattore di Conduzione k

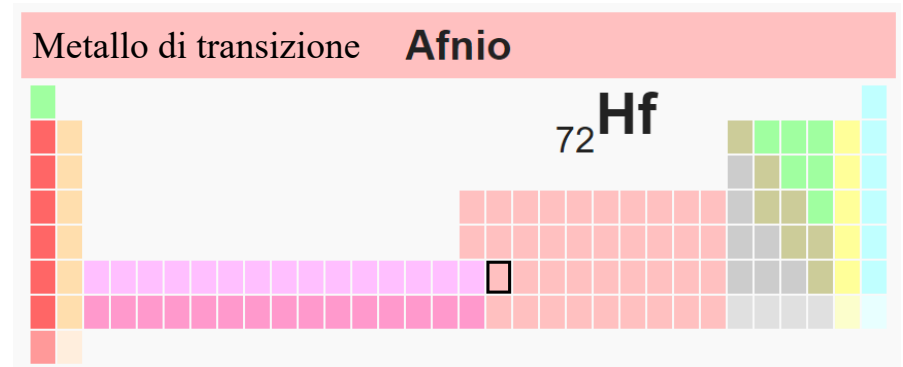
$$k_n = k'_n \frac{W}{L}$$

ϵ elevato (*high k*) da nodo a 45nm:

$$\epsilon_r(\text{diossido di silicio SiO}_2) = 3.9$$
$$\epsilon_r(\text{diossido di afnio HfO}_2) = \mathbf{25}$$

$$k'_n = \mu_n C_{ox} = \frac{\mu_n \epsilon_{ox}}{t_{ox}}$$

+ Gate Metallico contro strato svuotato del polisilicio (t_{ox} equiv. più grande)



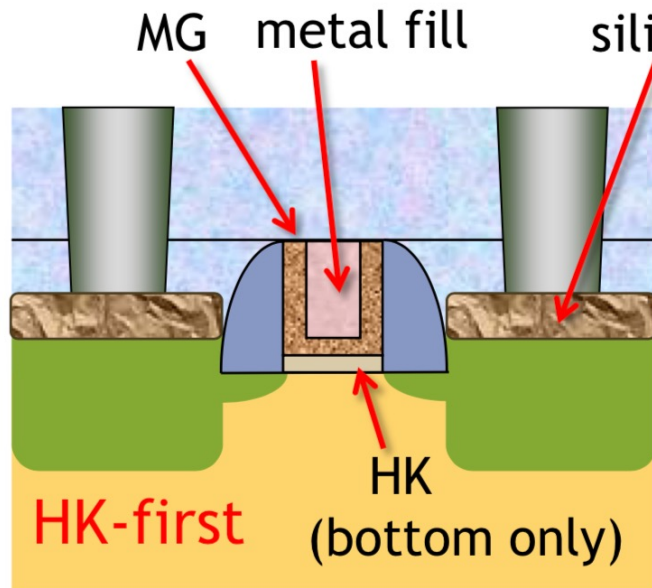
Strained Silicon

Minimo, limitato da corrente di *leakage* per effetto tunnel. Nodo a 65nm ha $t_{ox} \approx 1.6\text{-}2.3\text{nm}$ (caso limite 5 strati atomici)

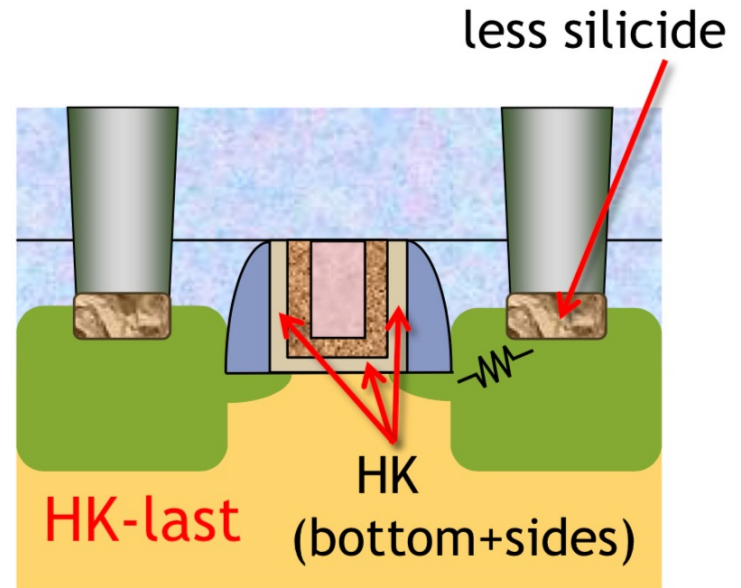
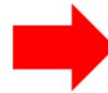
- Corrente di OFF nei MOS
- Limiti del Fattore di Conduzione k nei MOS
 - *High k*
 - *Strained Silicon*
- Fotolitografia

High k Metal Gate (HKMG)

- Higher C_{ox} without I_{gate} & poly depletion, but finicky interface
- Replace gate after S/D anneal for stable $V_T = f(\Phi_M)$
- Gate = (ALD MG stack to set Φ_M) + (metal fill to reduce R_G)
- Variation in MG grain orientation $\rightarrow V_T$ variation
- HK-first $\rightarrow$ HK-last for better gate edge control



Auth *et al.*, Intel



Packan *et al.*, Intel

Alvin L.S. Loke, 2016

High-k/Metal Gate Technology

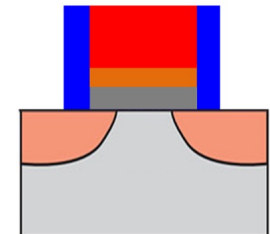
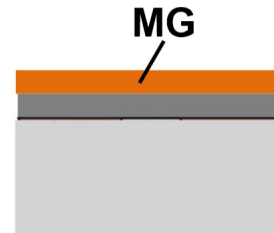
Nuo Xu, EE 290D, Berkeley, 2013

- | | |
|---|--|
|  SiN |  Poly-Si |
|  Low κ |  Metal Gate |
|  High κ |  W |
|  Si | |

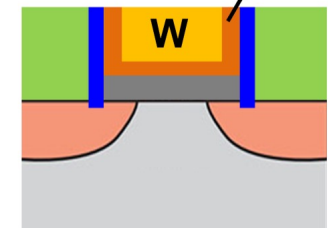
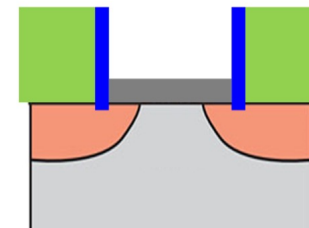
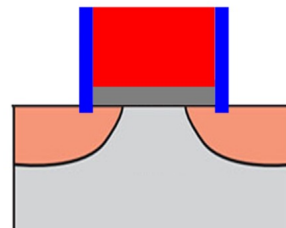
Process Flow



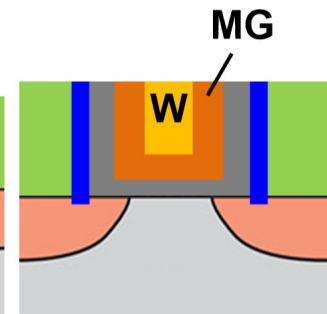
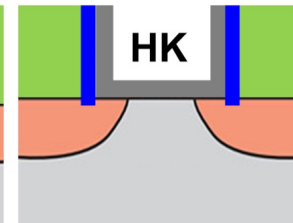
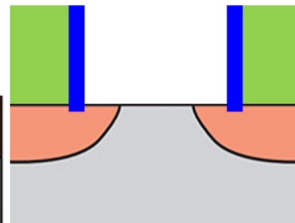
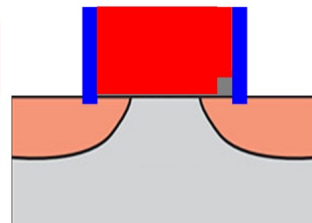
Gate First / MIPS
(Metal-Inserted-
Poly-Si Gate)



High- κ First

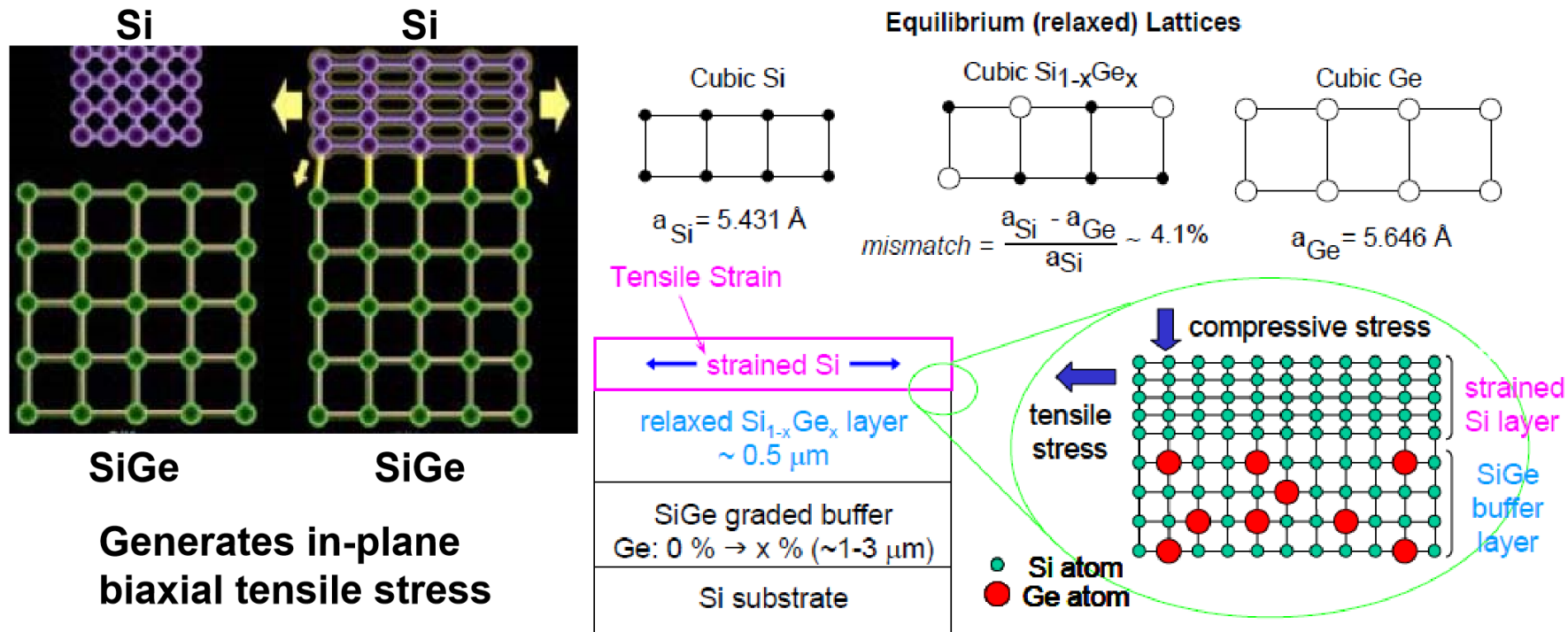


Gate Last / RMG
(Replaced Metal Gate)



- Corrente di OFF nei MOS
- Limiti del Fattore di Conduzione k nei MOS
 - *High k*
 - *Strained Silicon*
- Fotolitografia

Substrate-induced Strain: SiGe/Si



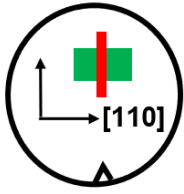
- In old generations of technology, strain is always introduced by growing Si thin channel on top of a relaxed SiGe layer.
- Drawbacks:
 - Enhanced substrate cost
 - Strain is global: for both P- and N- MOS
 - Thermal budget

10/25/2013

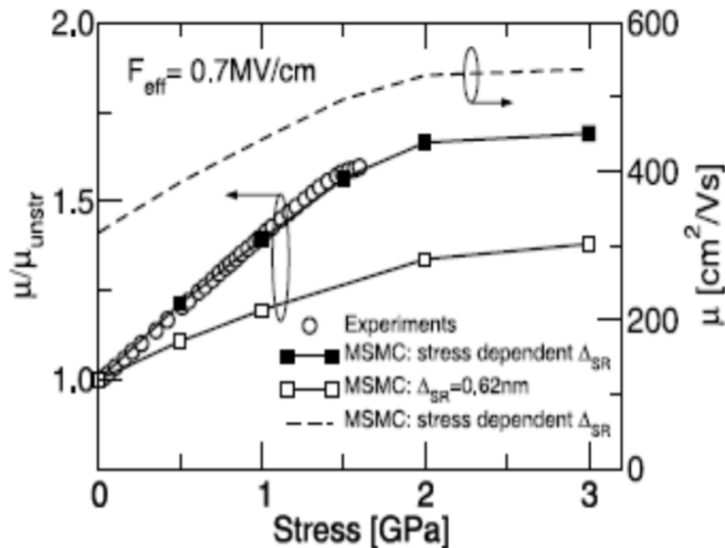
Nuo Xu

EE 290D, Fall 2013

Strain Enhancement on Si MOSFET Carrier Mobility

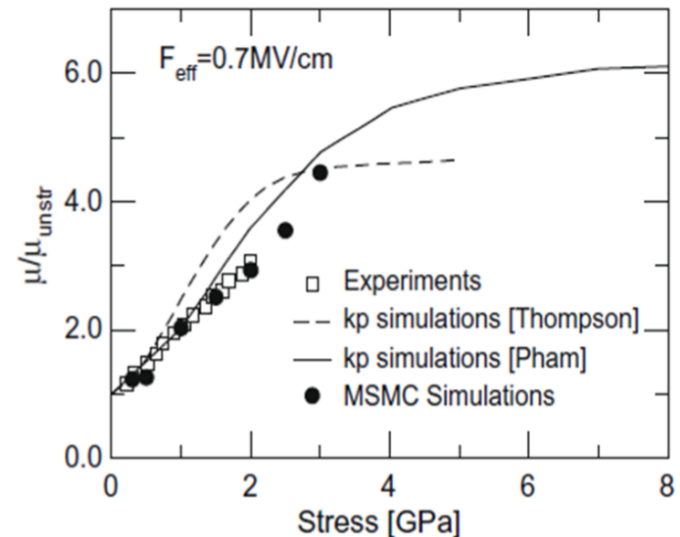


Electron



D. Esseni, JCE (2009)

Hole

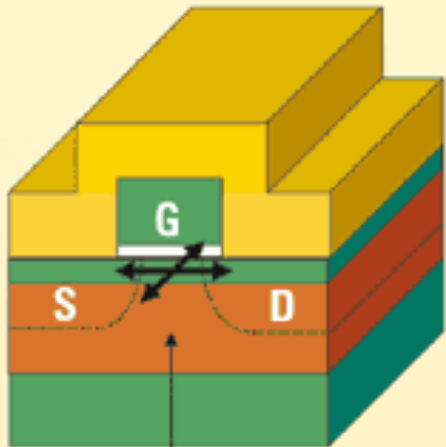


F. Conzati, SSE (2009)

- For N-MOSFET, electron mobility enhancement saturates at ~1.5 GPa $\langle 110 \rangle$ -uniaxial tensile stress .
- For P-MOSFET, hole mobility enhancement saturates at very high $\langle 110 \rangle$ -uniaxial compressive stress.

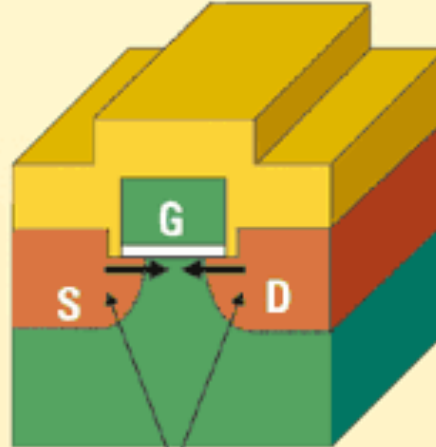
TRANSISTOR STRAIN TECHNIQUES

Traditional approach

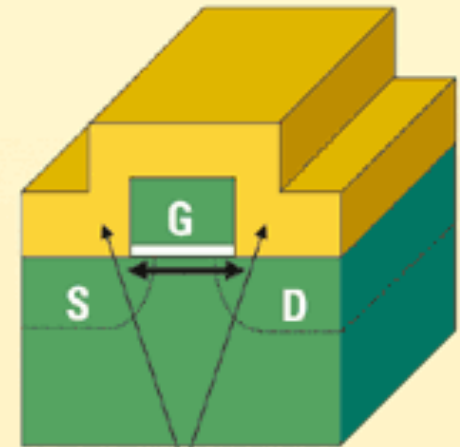


Graded SiGe layer
Biaxial tensile strain

Intel's 90 nm technology

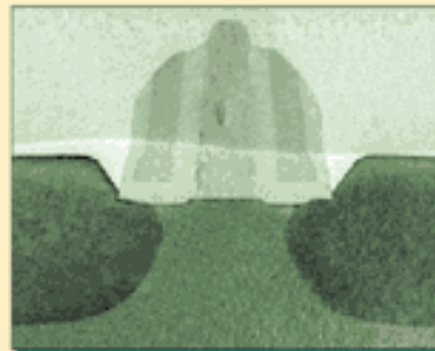


Selective SiGe S-D

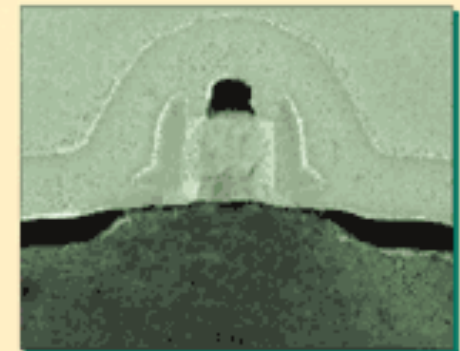


Tensile Si_3N_4 cap

Mobilità degli elettroni:
fra +50% e 70%
Mobilità delle lacune:
fra +300% e 500%
→ equalizzazione p-n



Uniaxial compressive
strain for PMOS



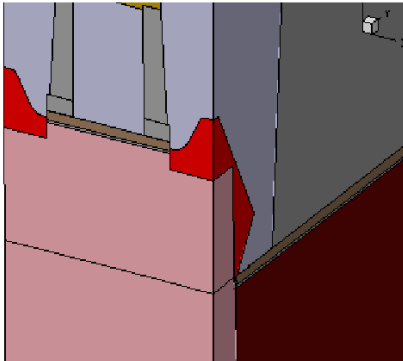
Uniaxial tensile
strain for NMOS

Intel

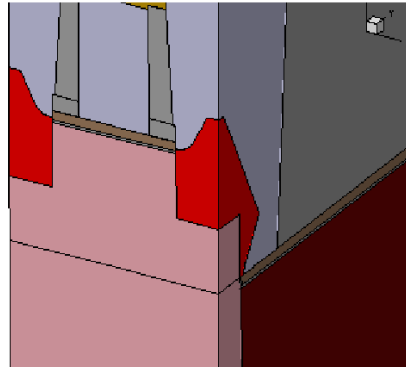
Impact of SiGe-S/D Strain on Fin Etch

Si_{0.5}Ge_{0.5} Source/Drain (Intel's 5th-gen strain Si tech.)

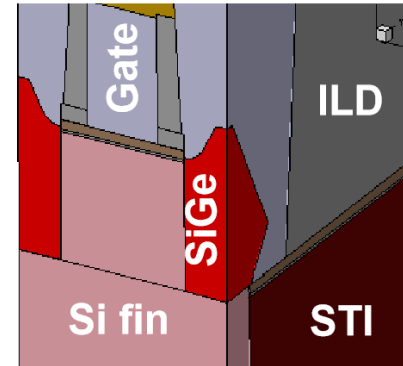
4 nm etch



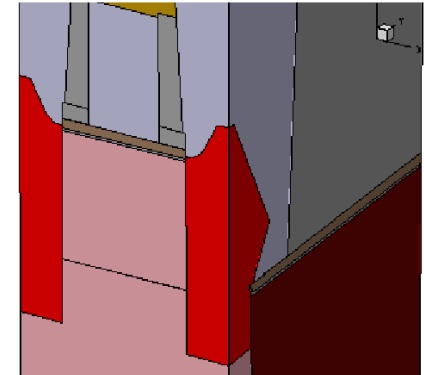
16 nm etch



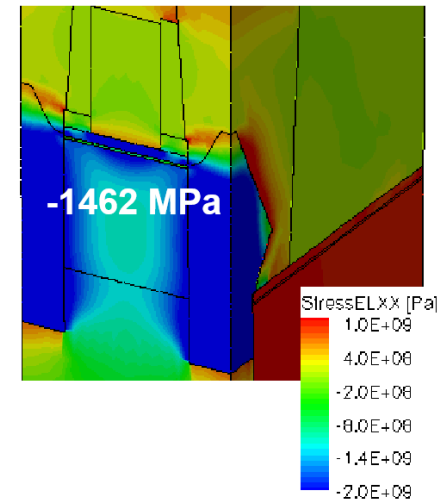
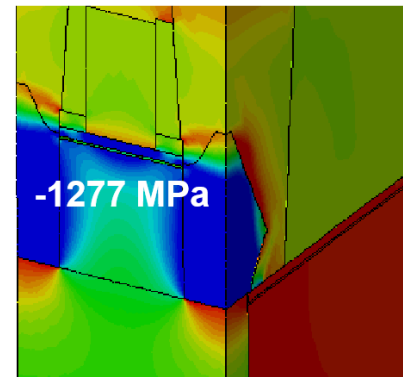
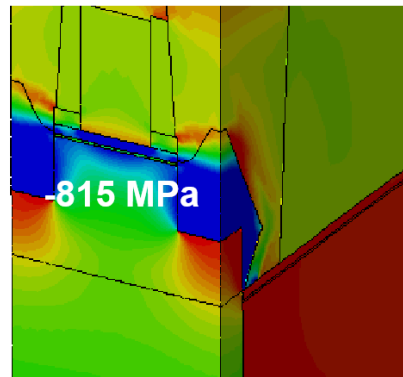
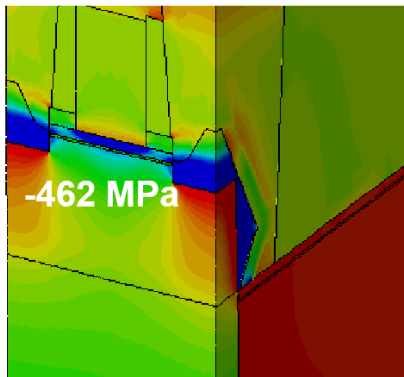
32 nm etch



48 nm etch



Longitudinal Stress (S_{xx})



- Deeper SiGe recess triples the channel stress!

M. Choi, ISTDM (2012)

10/25/2013

Nuo Xu

EE 290D, Fall 2013

Ci sono tante altre tecniche!

New FinFet Semiconductor Structure and Function

3:48

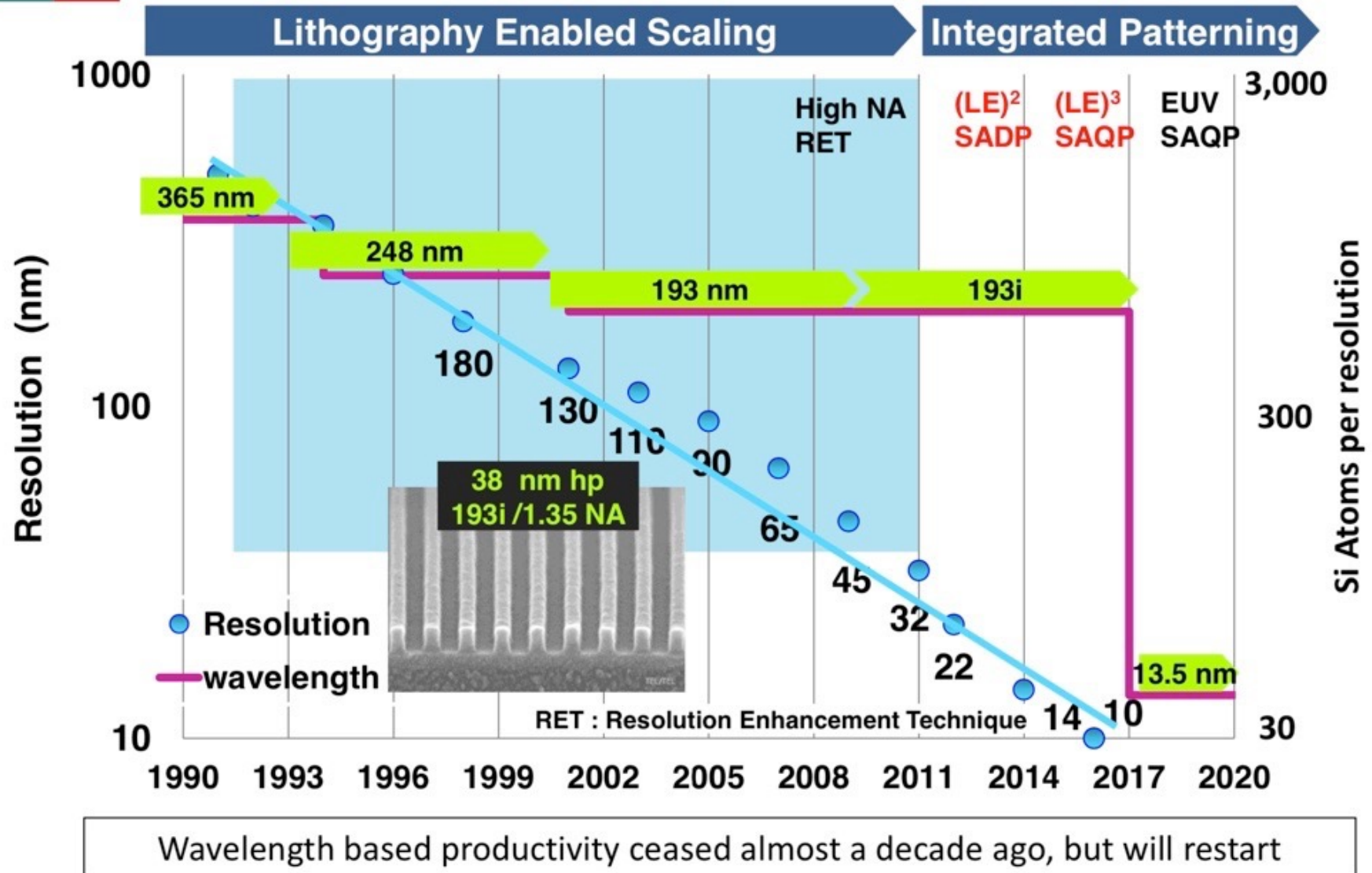
<https://youtu.be/c-3p8moNXfl>



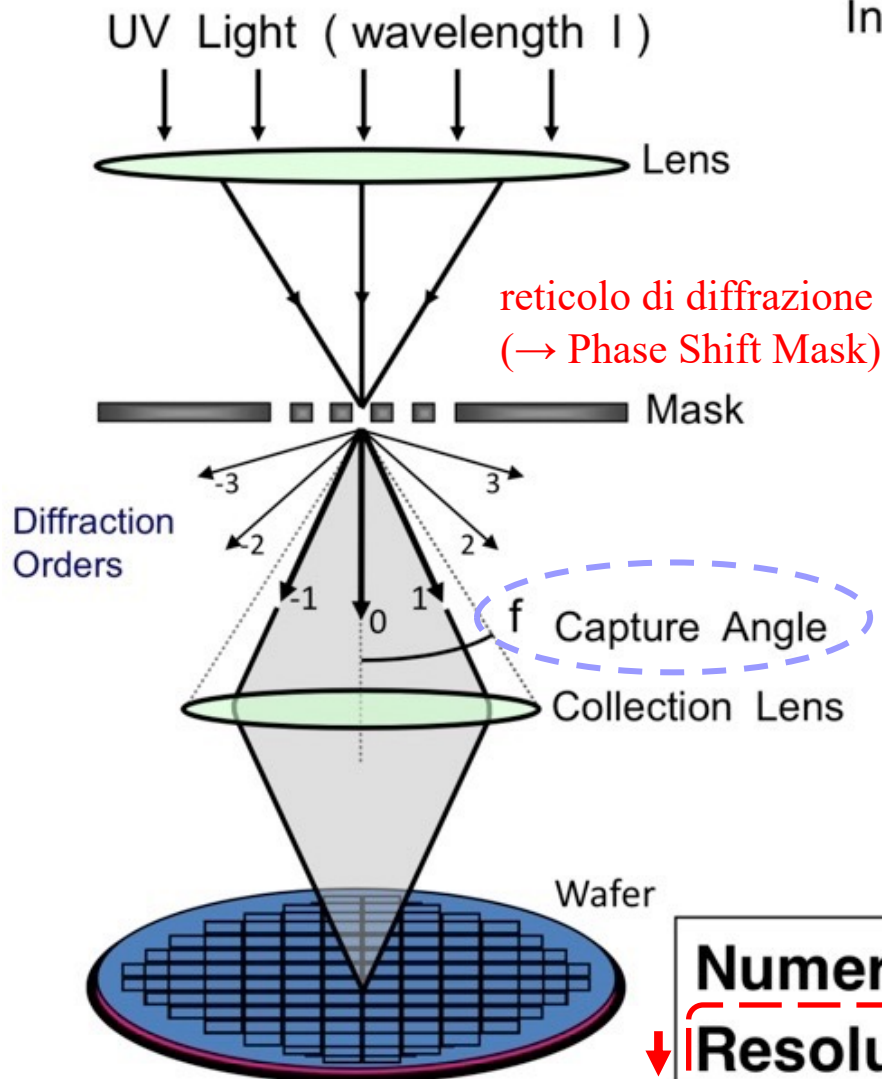
- Corrente di OFF nei MOS
- Limiti del Fattore di Condizione k nei MOS
- **Fotolitografia**

- • **ArF** – argon fluoride laser light : 193nm
- **CDU** – critical dimension uniformity
- **DOF** – depth of focus
- • **DPT** – double patterning technology
- • **EUV** – extreme ultraviolet light : 13.5nm
- **HP** – half pitch
- • **LELE** – litho/ etch/ litho/ etch double patterning
- **MEBDW** – massively parallel electron beam direct write
- **ML2** – maskless lithography
- • **NA** – numerical aperture
- **PV band** – Process Variation (through dose and focus) CD variance
- **PW** – exposure latitude vs depth of focus process window
- • **OPC** – optical proximity correction
- **QPT** – quadruple patterning
- • **SMO** – source mask optimization
- • **SPT** – spacer double patterning (aka **SADP** sidewall assisted DP)

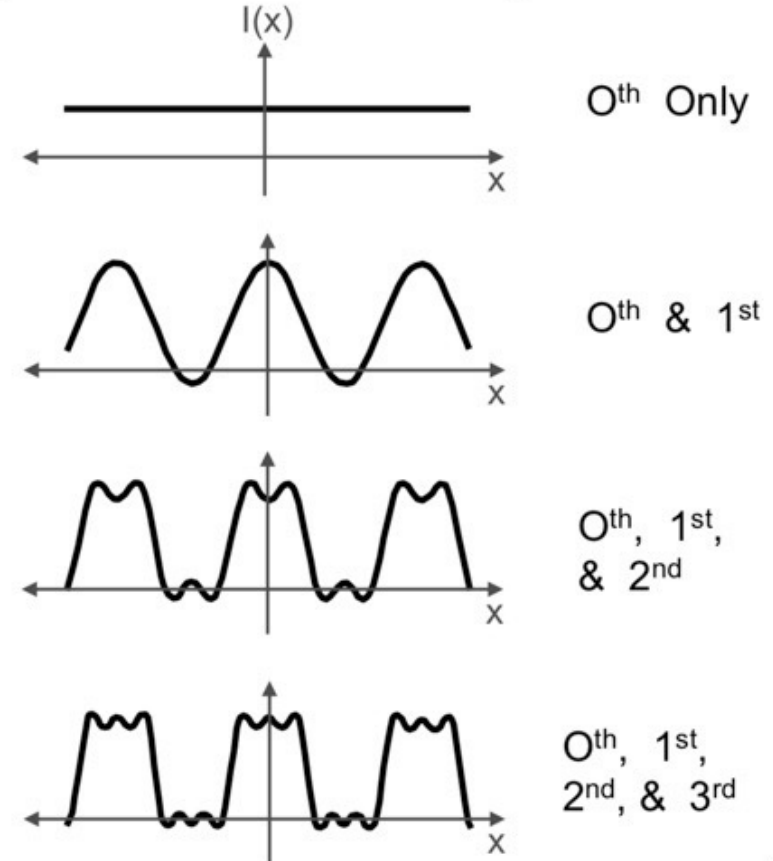
Lithography – Imaging History



Lithography – Imaging Principles



Intensity Profile vs. Orders Captured



Numerical Aperture: $\uparrow NA = \uparrow n \cdot \sin f \uparrow$

Resolution = $\downarrow k_1 * \downarrow \lambda / NA \uparrow$

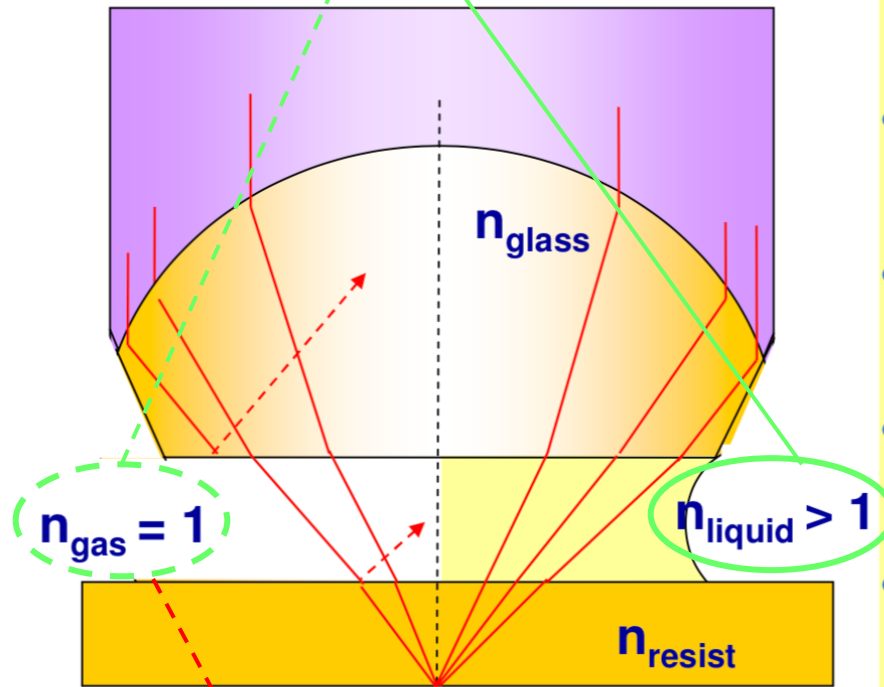
n : indice di rifrazione

f : apertura angolare della lente (1/2 cono)

k_1 : dipende dal processo (dal primo anello nero centrale nel pattern di diffrazione di Airy)

Increasing NA with immersion

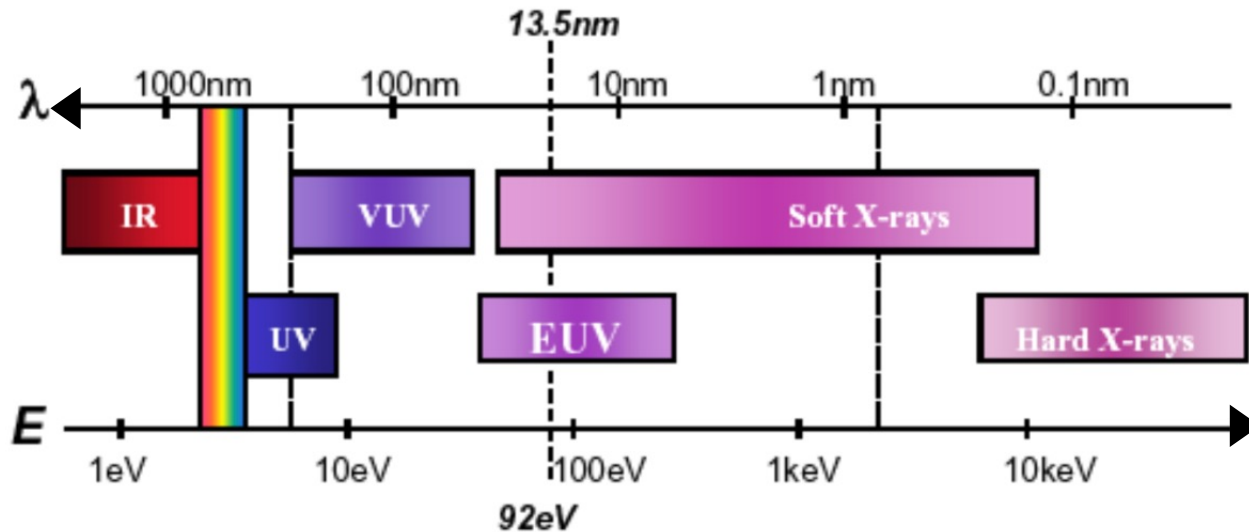
$$NA = n * \sin(\theta)$$



Limita anche θ per effetto Angolo Limite:
 $\theta_{limite} = \arcsin \frac{n_2}{n_1}$

- 1840: Oil Immersion optics
- 1877: Immersion Microscope
- 1980: 1st Immersion Lithography Patent
- 2000: Start ArF Immersion R&D
- 2003 First Immersion Scanner Images
- 2005 First Devices Made with Immersion
- 2011 Immersion is the Primary Lithography used for 40nm and Below

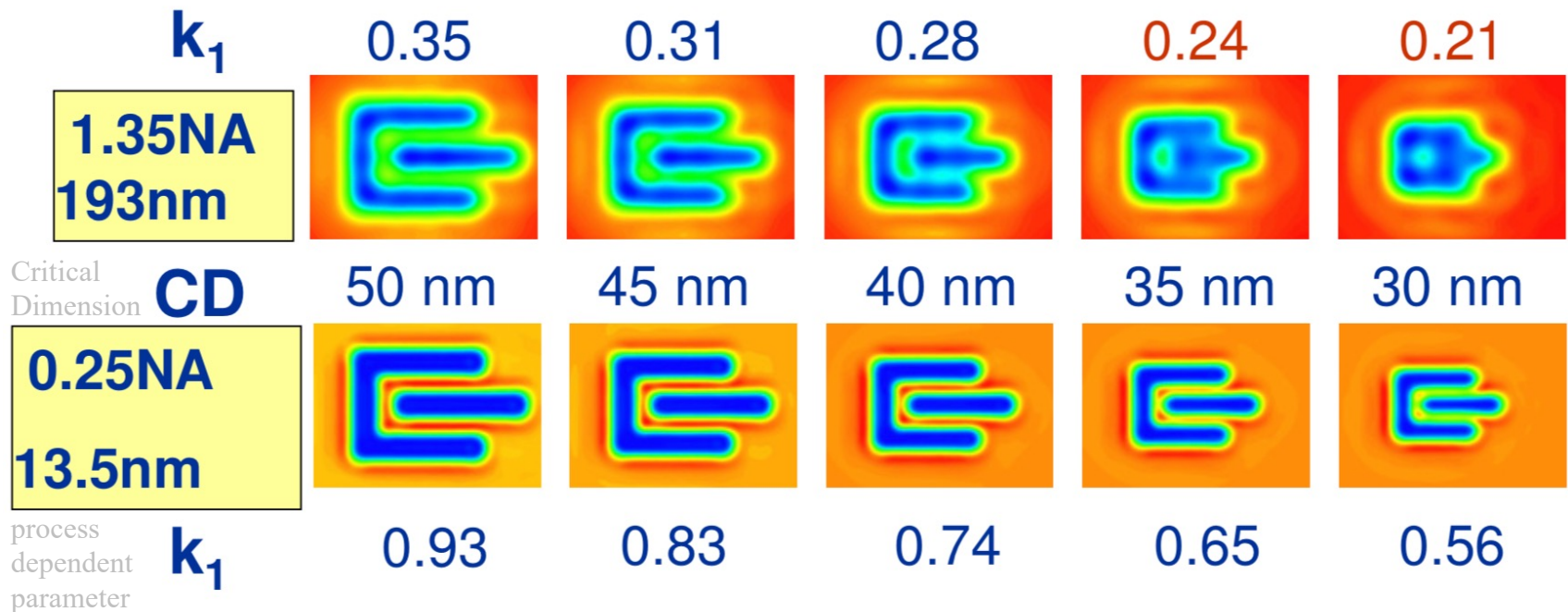
EUV Lithography (EUVL)



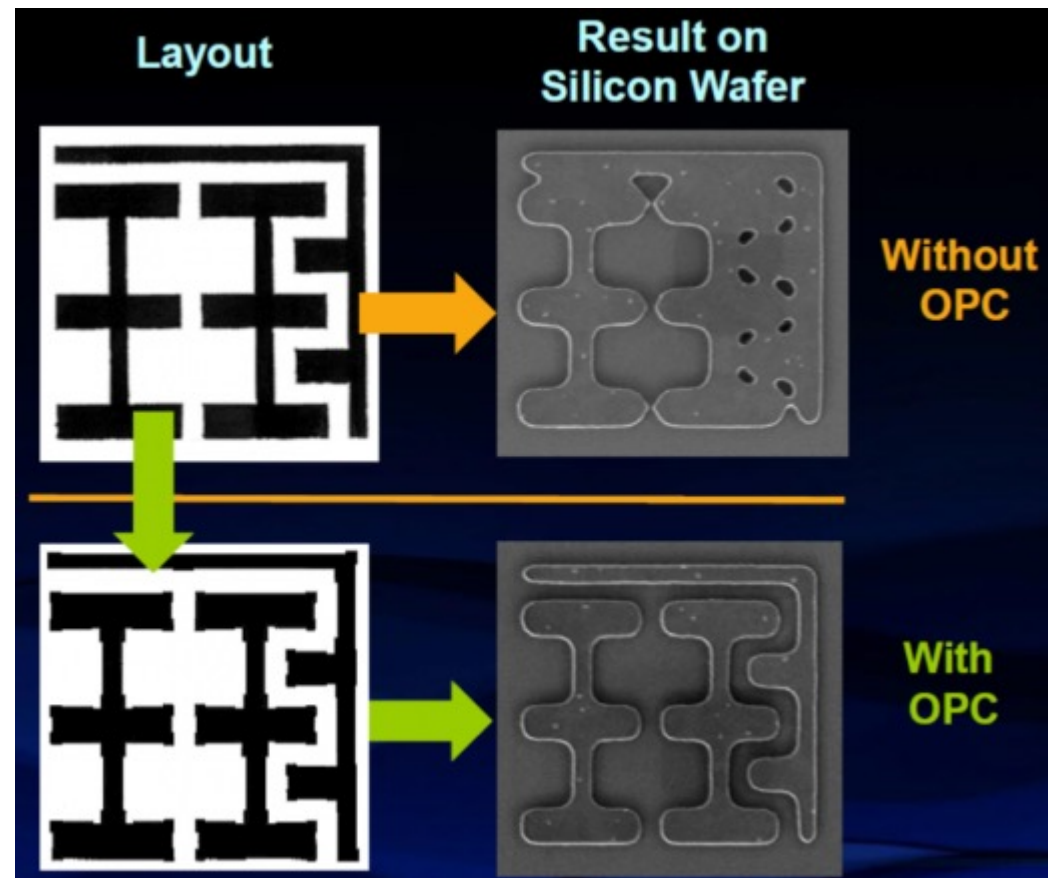
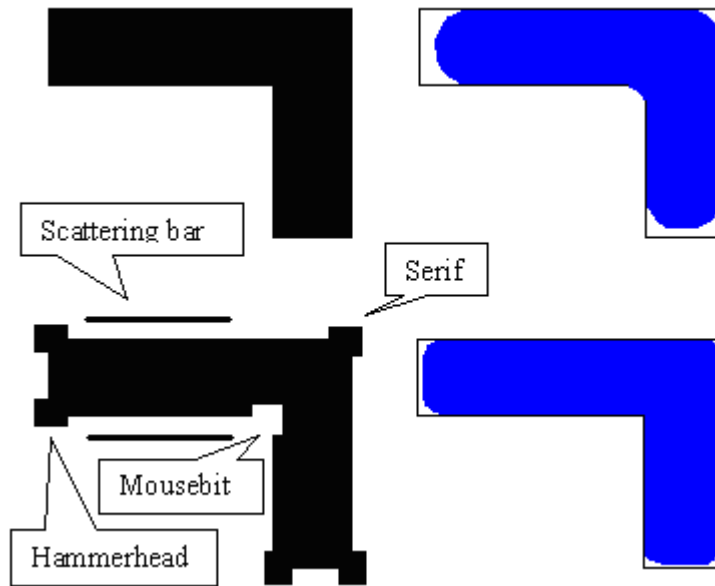
- Extreme UltraViolet (EUV) is EM **radiation at ~13.5 nm**, which falls into X-ray band. EUV was 'marketing name' to separate the program from hard X-ray lithography.
- EUV radiation is absorbed by almost all materials and gases, so the optics must be **reflective** and fully contained in **vacuum**.
- The reticle too must be reflective, **no pellicle** can be used to keep defects out of focus.

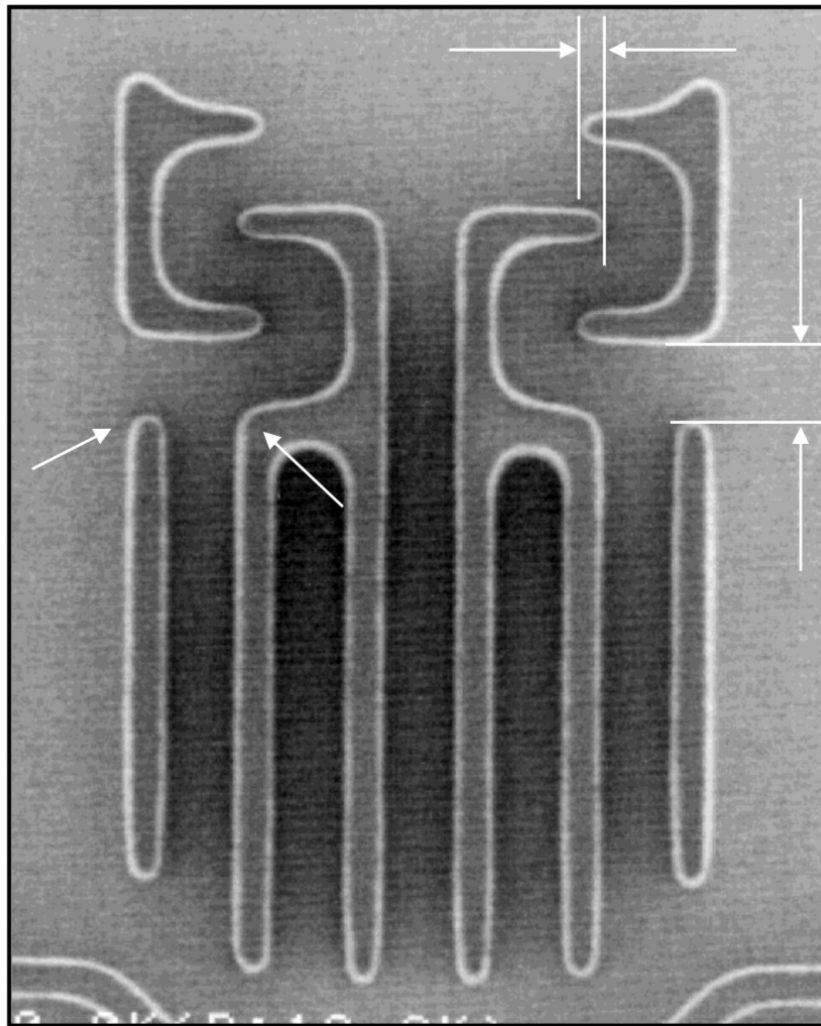
EUVL imaging has better fidelity than ArFi : high k1

- In the size range 50-10nm, 0.25 and 0.33 NA EUVL operates at much higher k1 than 1.35NA ArFi

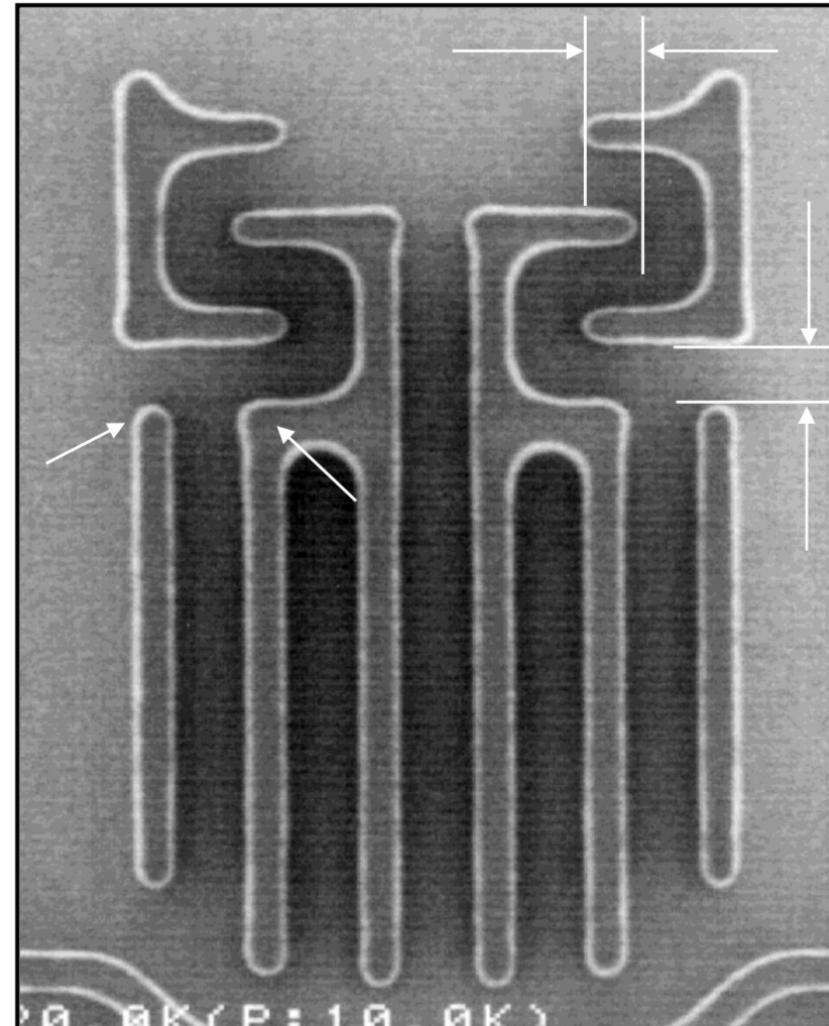


Optical Proximity Correction (OPC)





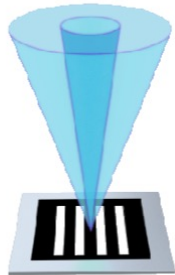
uncorrected



after OPC

More complex source shapes enable device shrinks

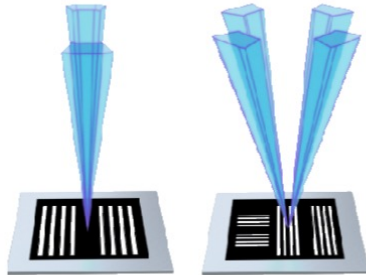
Lower k1 imaging requires increased pupil shape flexibility



early 90's

Aerial illuminator
with Zoom-Axicon

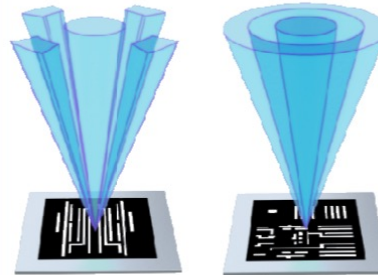
Conventional
& Annular



2000

DOE

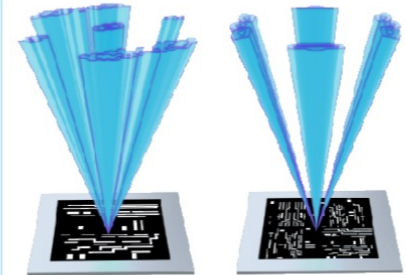
Dipole &
Quadrupole



2005 – 2008

DOE

Hexa-/more poles
Soft-pole, Multi-ring,
Asymmetric...



2009 – 2011

FlexRay

Freeform unlimited
Immediate shape variety

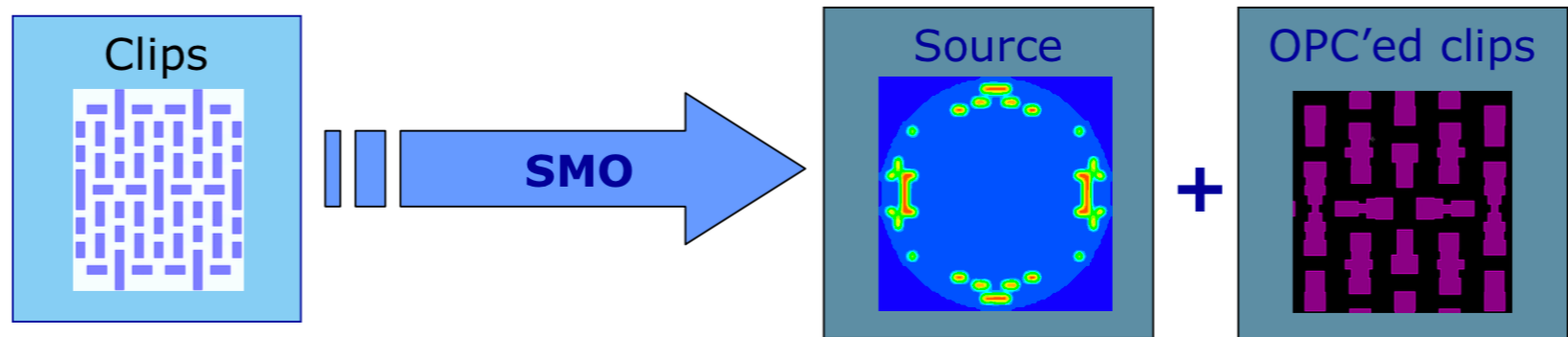
Programmable illumination (FlexRay) replaces fixed diffractive optical elements (DOE)

Source Mask Optimization (SMO)

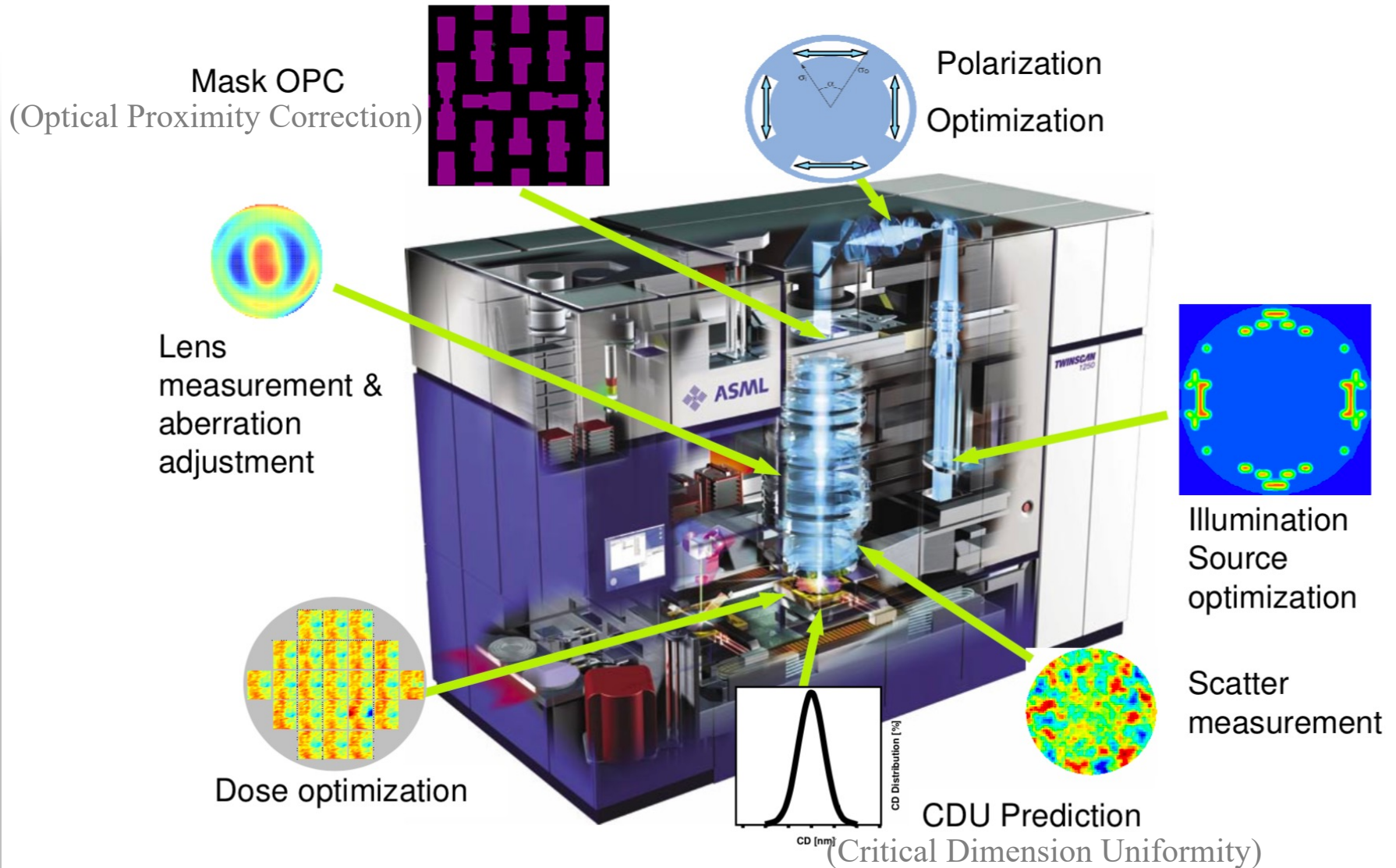
Source Mask Optimization (SMO)

What is Source Mask Optimization?

→ Co-optimizes illumination pupil shape (source) and OPC (mask) for critical design patterns



Immersion lithography extensions with mask and system holistic integration



William H Arnold

IEDM 2011 Short Course • Lithography for the 14nm Node and Beyond

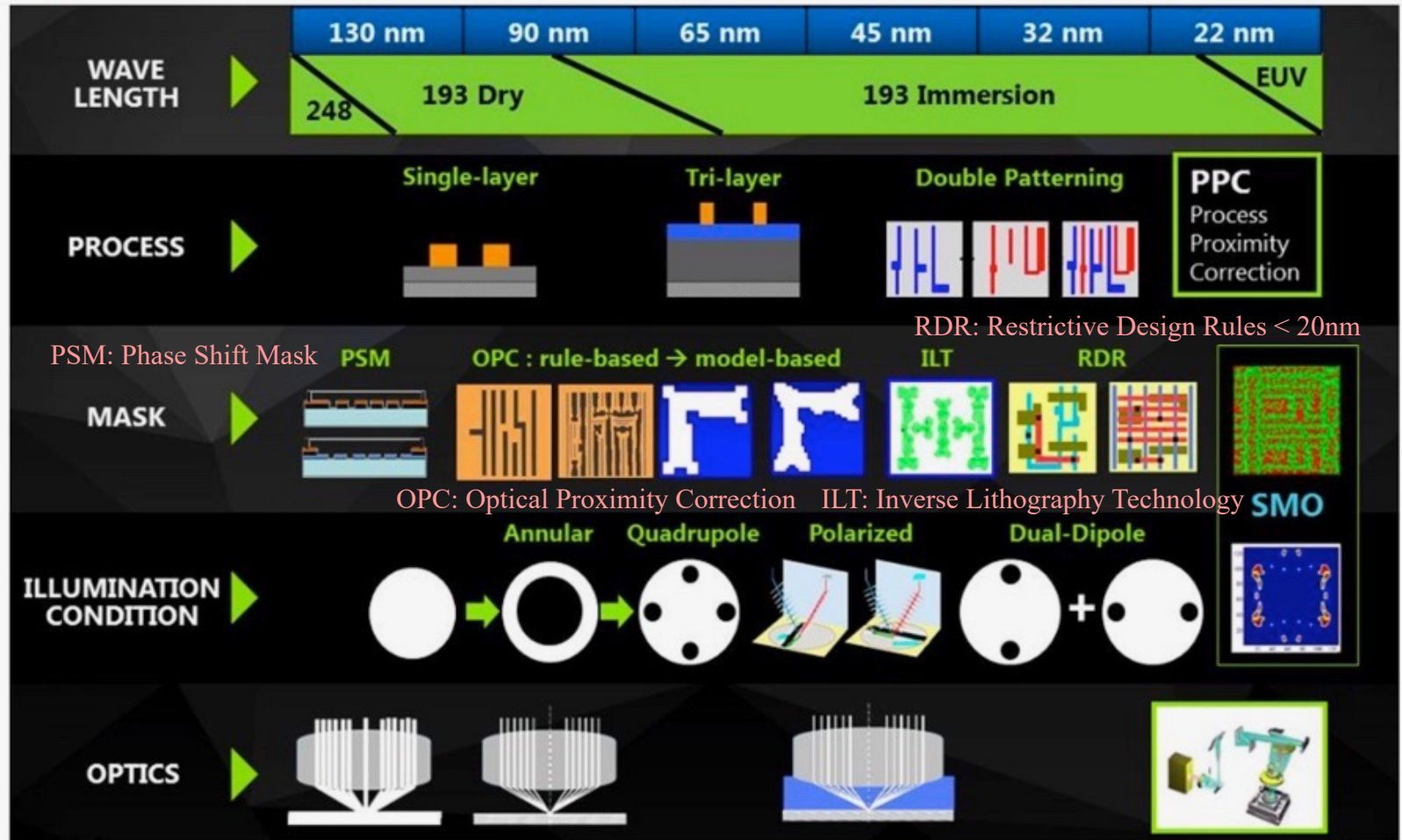
Extreme ultraviolet lithography

6:43

<https://youtu.be/6LfFjmWCWPw>



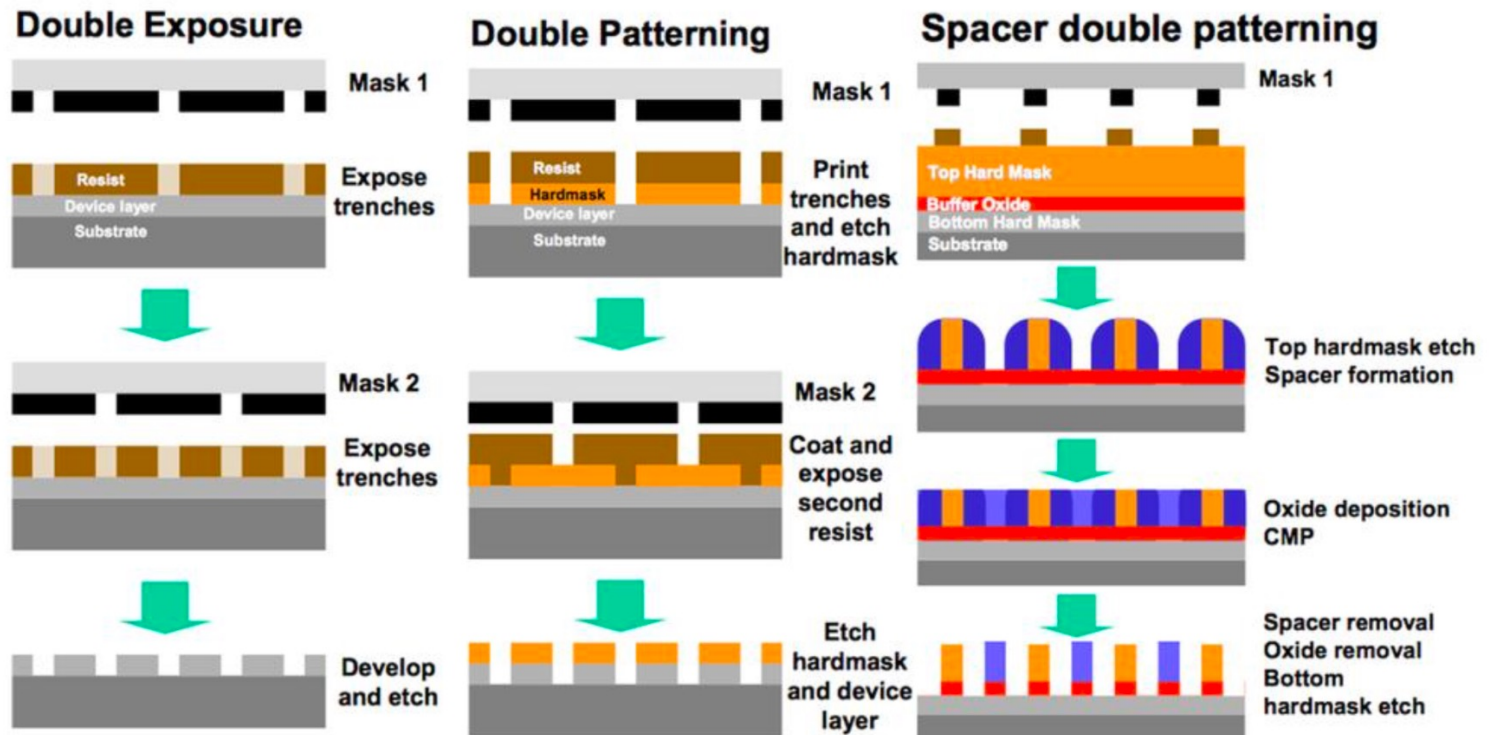
Lithography - Imaging



Wavelength only based productivity did cease almost a decade ago

Multi-Patterned Processing

- Solves lithography limitations by creating a single layer in multiple steps
 - Adjacent geometries created in separate steps are assigned different “colors”

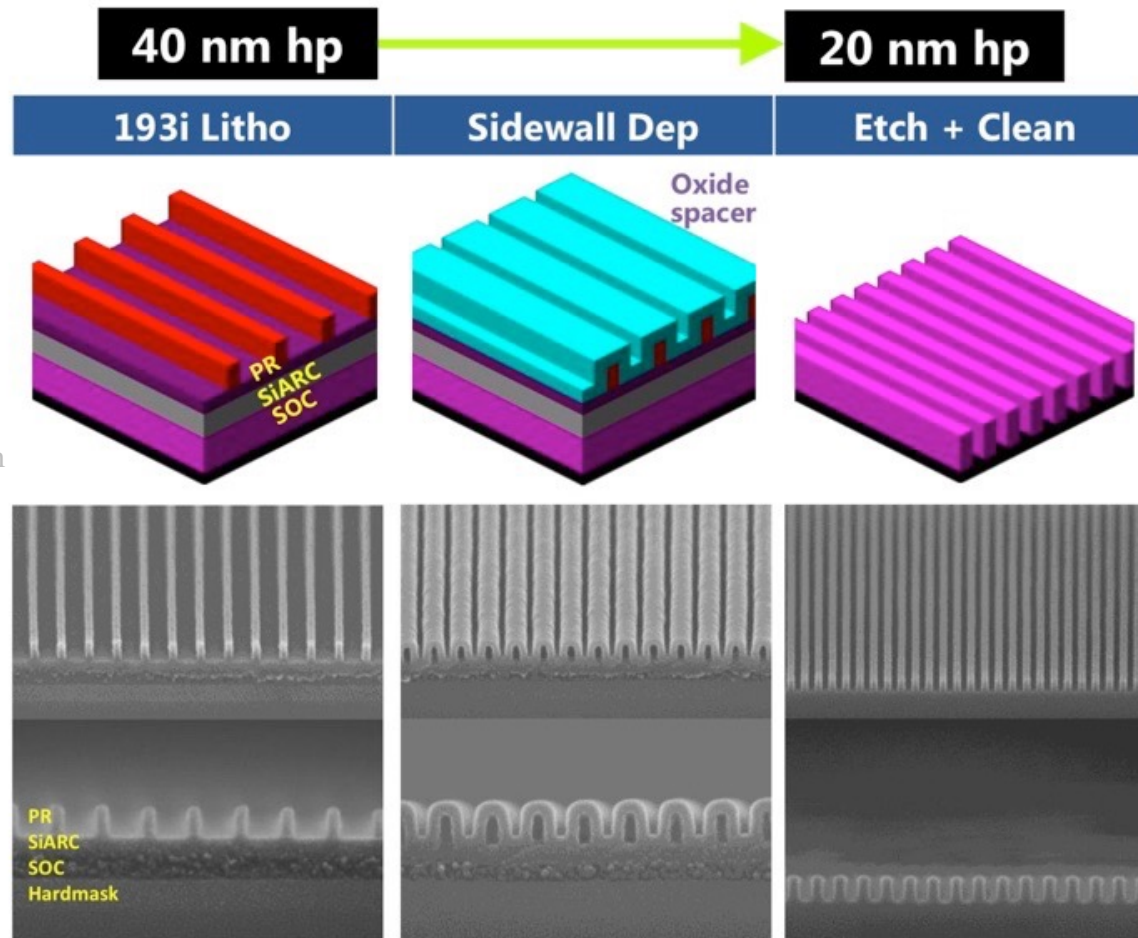


Self Aligned Double Patterning (SADP)

da Akihisa Sekiguchi, 2016



Self-Aligned-Double-Patterning



PR: Photo Resist

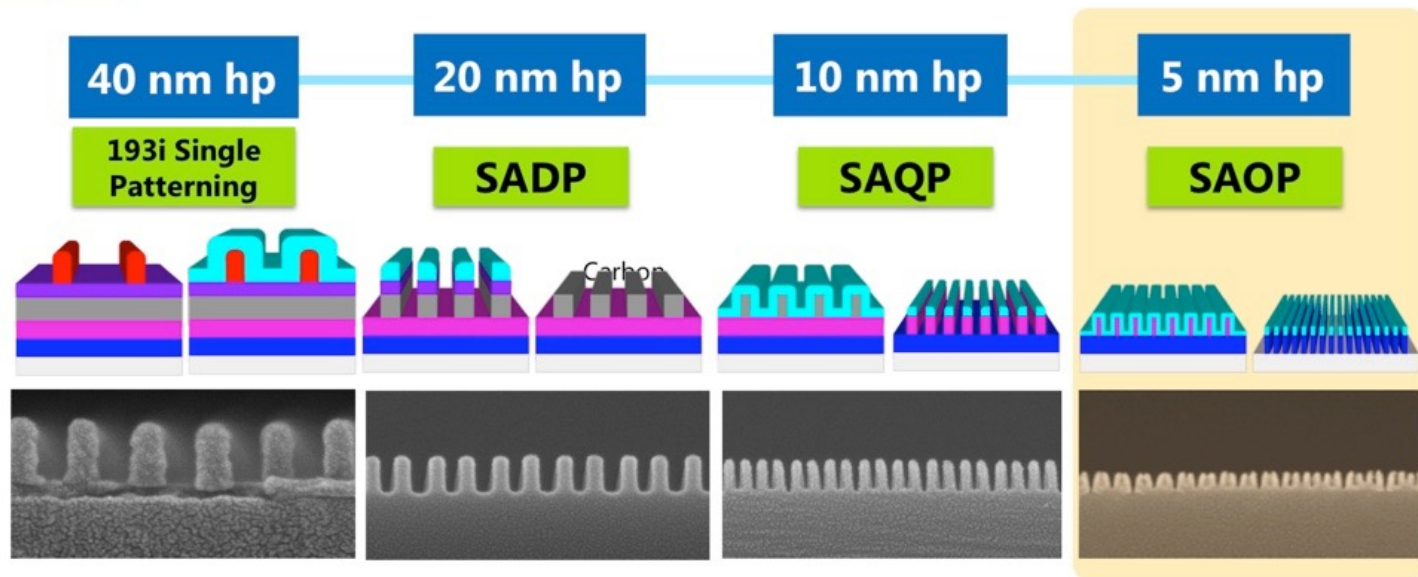
SiARC: Silicon Anti-Reflecting Coating

SOC: Spin-On Carbon

Key concepts are: self-alignment, overlay, deposition and etch process control



SADP Extendibility



SAOP : Self-Aligned Octuple Patterning

of process steps

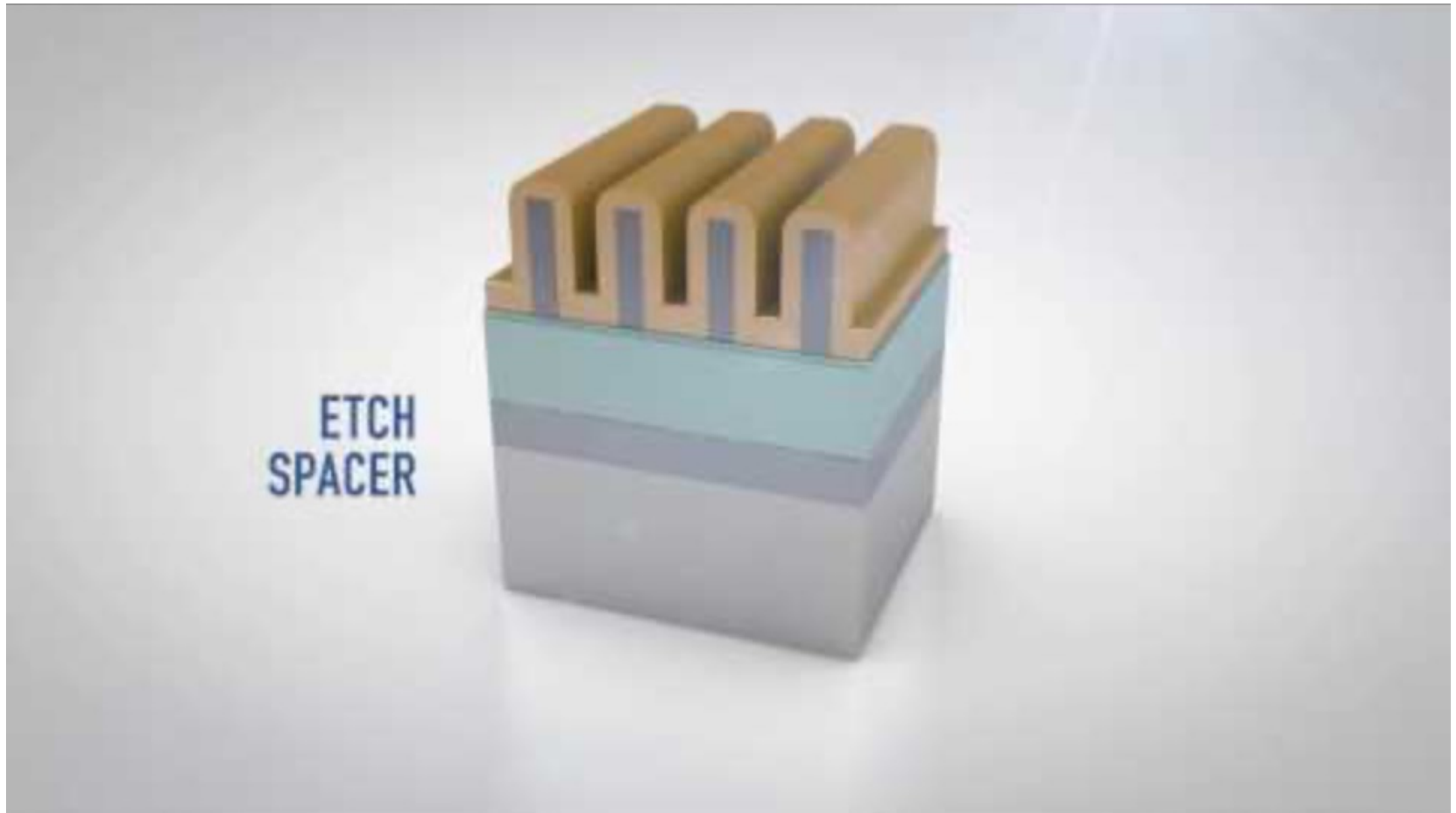
Lithography	1	2 = (core + cut)	4 = (core + cut) x 2	6 = (core + cut) x 3
Deposition	0	1 = (sidewall dep)	2 = (sidewall dep) x 2	4 = (sidewall dep) x 3
Etch	0	3 = (sw + cut + transfer)	6 = (sw + cut + transfer) x 2	9 = (sw + cut + transfer) x 3

Trade offs: lithography, self-alignment, overlay, deposition and etch process control, cost

Multiple Patterning Enables Feature Shrink

2:54

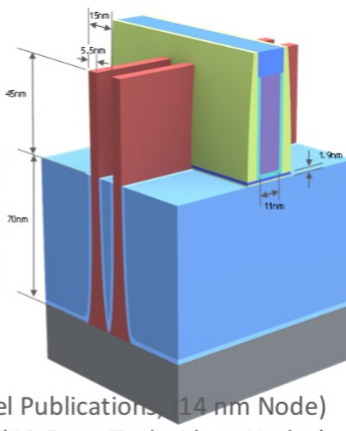
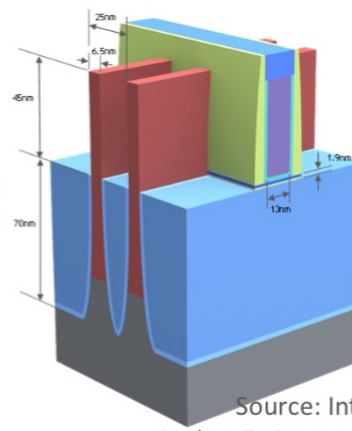
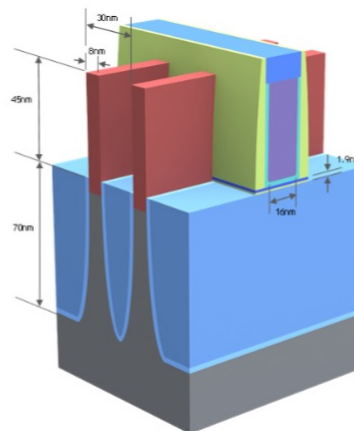
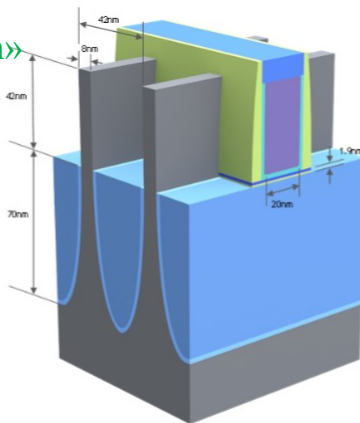
<https://www.youtube.com/watch?v=XI5Ypy77fVE>



Moore's Law

Year In Production	14 nm Node 3-D Tri-Gate (Published)	10 nm Node Band Engineered 3-D Tall Tri-Gate (Estimates)	7 nm Node Band Engineered 3-D Tall Tri-Gate (Estimates)	5 nm Node Band Engineered 3-D Tall Tri-Gate (Estimates)
Lithography Node	14 nm	10 nm	7 nm	5 nm
Gate Length	L = 20 nm (Z = 85 nm) (L = 22-24 nm per chipworks)	L = 16 nm	L = 13 nm	L = 11 nm
Lithography	193nm ArF Immersion + Pitch Division Spacer Patterning (Pitch Halving)	193nm ArF Immersion + Pitch Division Spacer Patterning (Pitch Quartering)	193nm ArF Immersion + Pitch Division + EUV Cuts and Vias	193nm ArF Immersion + Pitch Division + EUV Cuts and Vias
Gate Dielectric Tox (Physical)	0.8nm Oxinitride+ 1.1nm High- κ_4	0.8nm Oxinitride+ 1.1nm High- κ_5	0.8nm Oxinitride+ 1.1nm High- κ_6	0.8nm Oxinitride+ 1.1nm High- κ_7
Gate Dielectric Tox_{inv} (Electrical/CET)	HKMG 0.9nm EOT PMOS: TiN 1.4nm / TaN 0.5nm / TiN 3.0nm / TiAlN Fill NMOS: TiAl 2.6nm / TiAlN 1.4nm / W Fill	HKMG 0.8nm EOT	HKMG 0.8nm EOT	HKMG 0.7nm EOT
Poly Pitch/Thickness	Metal 70/37nm (Fin Pitch 42nm, 32nm Recessed Gate)	Metal 52/25nm	Metal 43/25nm	Metal 33/25nm
Isolation Pitch/Depth	Pitch: 42nm Depth: 104nm, 115nm, 125nm, 137nm From Fin Top	Pitch: 40nm Depth: 125nm (Mean) From Fin Top	Pitch: 40nm Depth: 125nm (Mean) From Fin Top	-
Contacted Gate Pitch	Vertical & Tall: Fin W _{Si} = 8nm Vertical & Tall: Fin H _{Si} = 42nm Vertical & Tall: Si Fin Pitch = 42nm Contacted Gate Pitch = 70nm	Vertical & Tall: Fin W _{SiGe} = 8.0nm Vertical & Tall: Fin H _{SiGe} = 45nm Vertical & Tall: SiGe Fin Pitch = 30nm Contacted Gate Pitch = 50nm	Vertical & Tall: Fin W _{SiGe} = 6.5nm Vertical & Tall: Fin H _{SiGe} = 45nm Vertical & Tall: SiGe Fin Pitch = 25nm Contacted Gate Pitch = 40nm	Vertical & Tall: Fin W _{SiGe} = 5.5nm Vertical & Tall: Fin H _{SiGe} = 45nm Vertical & Tall: SiGe Fin Pitch = 15nm Contacted Gate Pitch = 30nm
Contact CD in Contacted Gate Pitch	Ti/TiN/W, Pitch = 70nm Bottom PMOS/NMOS CD = 22nm Bottom/Side Ti+TiN = 7nm/4nm	Bottom PMOS/NMOS CD = 16nm	Bottom PMOS/NMOS CD = 16nm	Bottom PMOS/NMOS CD = 12nm
Metal 0 Pitch/Thick. - AR/PR Liner Thickness: SW/Bottom Via0 Pitch/Depth ILD0 Material / Thickness (nm)	W/TiN 71/65nm - 1.6/1.4x SW: [Ti/TiN 3/3nm], Bottom [Ti/TiN 5/5nm] Via0: 65/38nm ILD0: T-SiOC/SiNO 45-56nm	W/TiN 45/40nm - 1.5/1.0x SW: [Ti/TiN 3/3nm], Bottom [Ti/TiN 5/5nm] Via0: x/30nm ILD0: T-SiOC/SiNO	Option A: Extend W with Modified Barriers Option B: Cobalt Alloys, In Development Option B: Cobalt Alloys, In Development Option B: Cobalt Alloys, In Development ILD0: T-SiOC/SiNO	Cobalt Alloys Cobalt Alloys Cobalt Alloys Cobalt Alloys ILD0: T-SiOC/SiNO

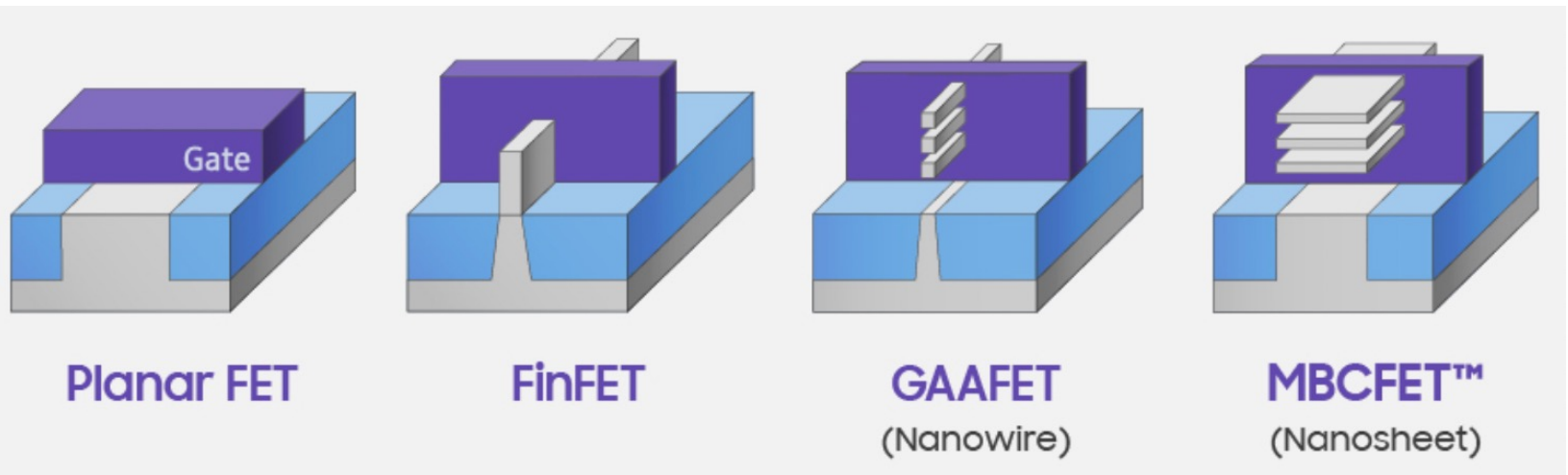
La tecnologia in «nm» oggi NON indica più la lunghezza minima di canale, 1/2 metal pitch e nemmeno le geometrie minime realizzabili...



Reza Arghavani
Challenges of 5nm CMOS Technology, 112

Source: Intel Publications, (14 nm Node)
Author Estimates, (10-5 nm Technology Nodes)

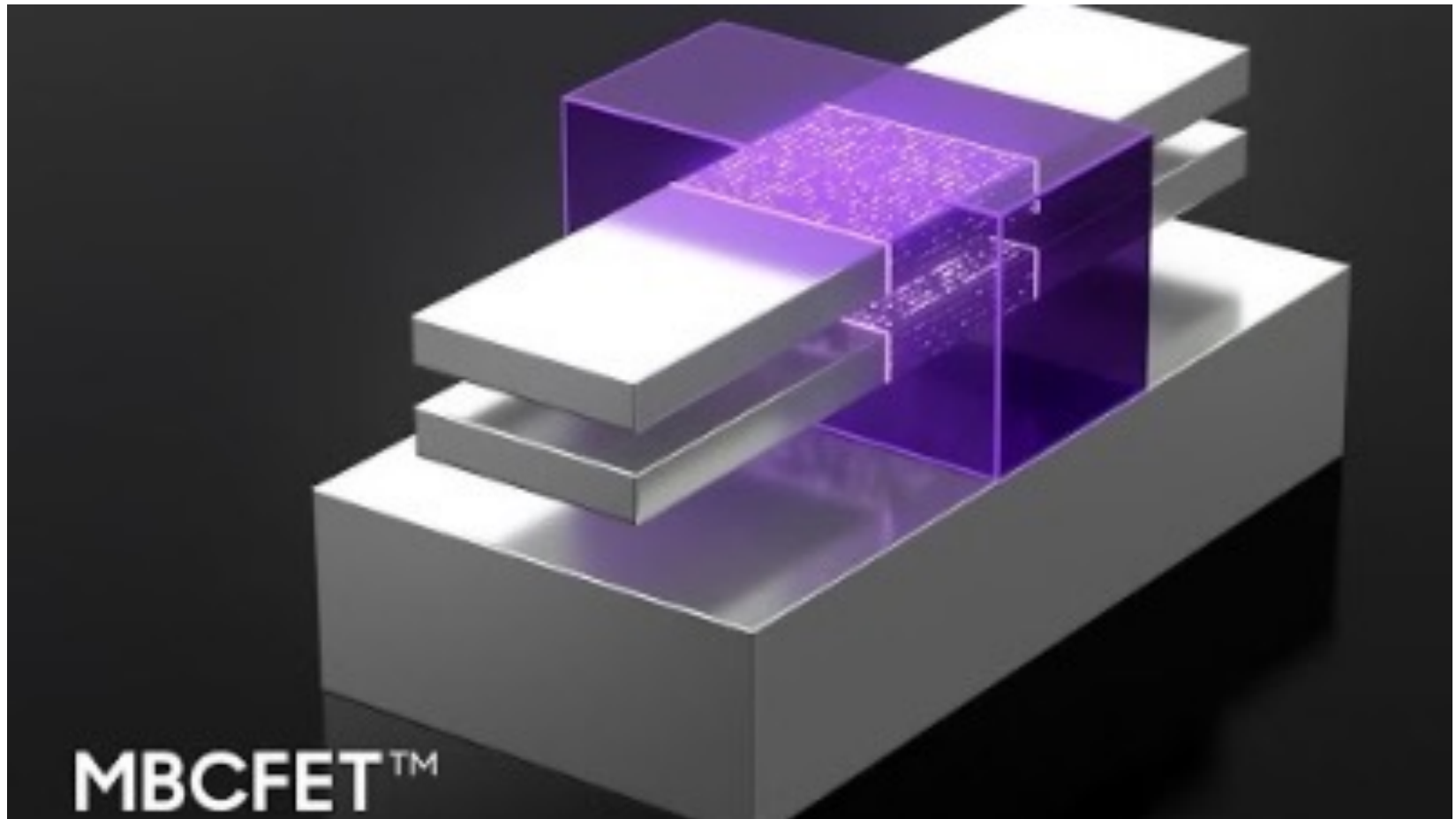
Ulteriore Evoluzione MOSFET



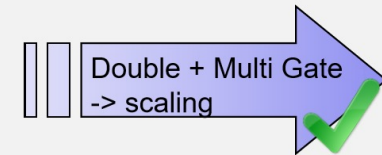
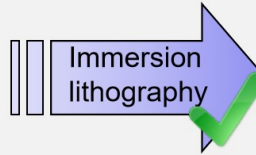
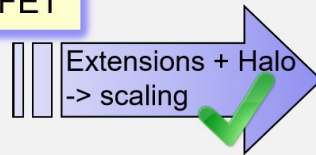
NANOSHEET (Samsung MBCFET)

2:05

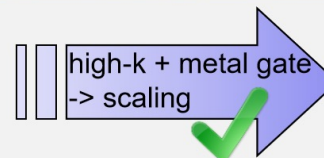
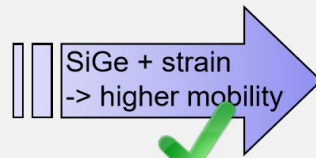
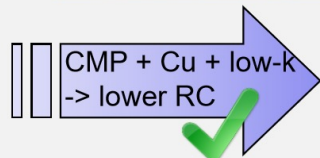
<https://www.youtube.com/watch?v=3otqUu-7WUQ>



Planar MOSFET

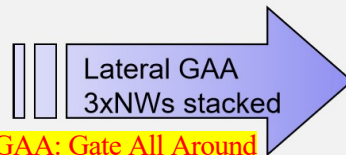


180 nm node 140nm – 100nm 1999 - 2001	130 nm node 90nm – 70nm 2002 - 2004	100/90 nm node 65nm – 50nm 2004 - 2006	70/65 nm node 45nm – 35nm 2006 - 2008	50/45 nm node 30nm – 25nm 2008 - 2010	35/32 nm node 20nm – 17nm 2010 - 2012	22 nm node 14nm – 10nm 2012 - 2014
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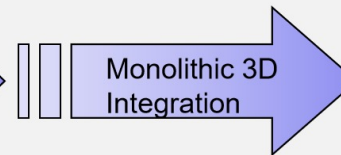


CMP: Chemical Mechanical Polishing

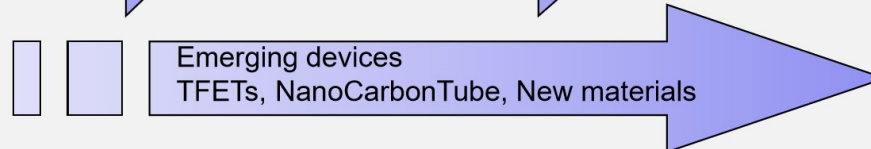
FinFET



GAA: Gate All Around



16/14 nm node 24nm 2014 - 2016	11/10 nm node 18nm 2016 - 2018	8/7 nm node 14nm 2019 - 2020	6/5 nm node 10nm 2021 - 2023	4/3 nm node 10nm 2024 - 2026	3/2.5 nm node 10nm 2027 - 2029	2/1.5 nm node 10nm 2030 -
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NANOSHEET (Intel RibbonFET + POWERVIA)

2:45

<https://www.youtube.com/watch?v=t6Y41zdO3Pc>



Riassunto dell'Evoluzione dei MOS

5:03

<https://www.youtube.com/watch?v=Z7M8etXUEUU>

